Q1

The potential energy of a particle under a conservative force is given by $U(x)=\left(x^2-3x\right)$ J. The equilibrium position of the particle is at

(1) $x=1.5$ m

(2) $x=2$ m

(3) $x=2.5$ m

(4) $x=3$ m

Q2

Two small satellites are moving in circular orbits around the earth at a distance R and R+ $\triangle$ R from the centre of the earth. If their time period of rotation are T and T+ $\triangle$ T respectively, then $(\Delta \mathrm{R}<\mathrm{K})$

(1) $\Delta \mathrm{T}=\mathrm{T} \frac{\Delta \mathrm{R}}{\mathrm{R}}$

(2) $\Delta \mathrm{T}=3 \mathrm{T} \frac{\Delta \mathrm{R}}{\mathrm{R}}$

(3) $\Delta \mathrm{T}=\frac{3}{2} \mathrm{T} \frac{\Delta \mathrm{R}}{\mathrm{R}}$

(4) $\Delta \mathrm{T}=\frac{2}{3} \mathrm{T} \frac{\Delta \mathrm{R}}{\mathrm{R}}$

Q3

Four resistance of 3 Ω, 3 Ω, 3 Ω and 4 Ω respectively are used to form a Wheatstone bridge. The 4 Ω resistance is short-circuited with a resistance R in order to get bridge balanced. The value of R will be

(1) 10 Ω

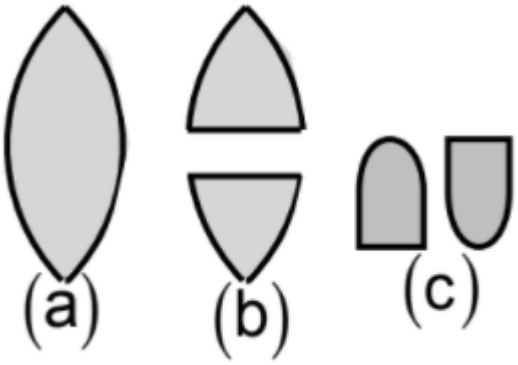
(2) 11 Ω

(3) 12 Ω

(4) 13 Ω

Q4

An equiconvex glass lens lens (a) has a focal length f and power P. It is cut into two symmetrical halves (b) by a plane containing the principal axis. The two pieces are recombined as shown in figure (c). The power of new combination is



(1) P

(2) $P/2$

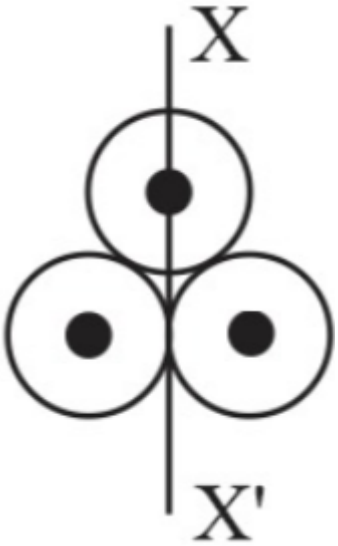
(3) $2P$

(4) zero

Q5

Three identical spherical shells, each of mass m and radius r are placed as shown in the figure. Consider an axis $\mathrm{XX}^{'}$ which is touching the two shells and passing through diameter of the third shell. Moment of inertia

of the system consisting of these three spherical shells about $XX^{'}$ axis is:



(1) $\frac{11}{5}mr^2$

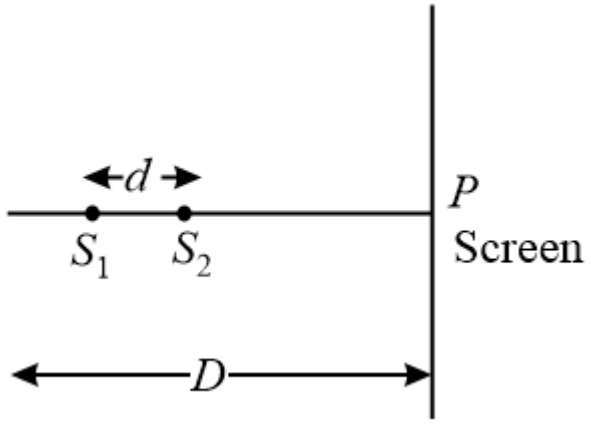
(2) $3\ mr^2$

(3) $\frac{16}{5}mr^2$

(4) $4\ mr^2$

Q6

Two coherent point sources S_1 and S_2 are separated by a small distance d as shown in the figure. The fringes obtained on the screen will be



(1) semi-circles

(2) concentric circles

(3) points

(4) straight lines

Q7

A sample of radioactive material decays simultaneously by two processes A and B with half-lives $\frac{1}{2}$ and $\frac{1}{4}$ hours respectively. For first half-hour, it decays with process A, next one hour with process B and for a further half an hour with both A and B. If originally there were N_0 nuclei, find the number of nuclei after 2 hours of such decay-

(1) $\frac{N_0}{(2)^8}$

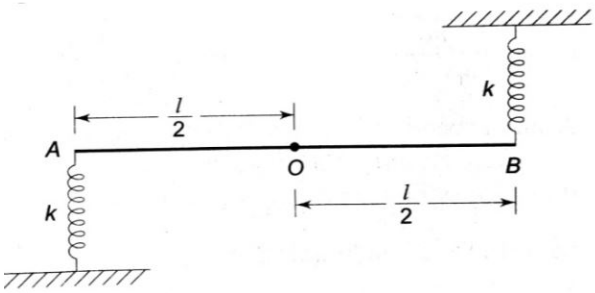
(2) $\frac{N_0}{(2)^4}$

(3) $\frac{N_0}{(2)^6}$

(4) $\frac{N_0}{(2)^5}$

Q8

The ends of a rod of length l and mass m are attached to two identical springs as shown in the figure. The rod is free to rotate about its centre O. The rod is depressed slightly at end A and released. The time period of the oscillation is



(1) $2\pi\sqrt{\frac{m}{2k}}$

(2) $2\pi\sqrt{\frac{2m}{k}}$

(3) $\pi\sqrt{\frac{2m}{3k}}$

(4) $\pi\sqrt{\frac{3m}{2k}}$

Q9

Assertion: The comets do not obey Kepler's laws of planetary motion.

Reason: The comets do not revolve in elliptical orbits.

(1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.

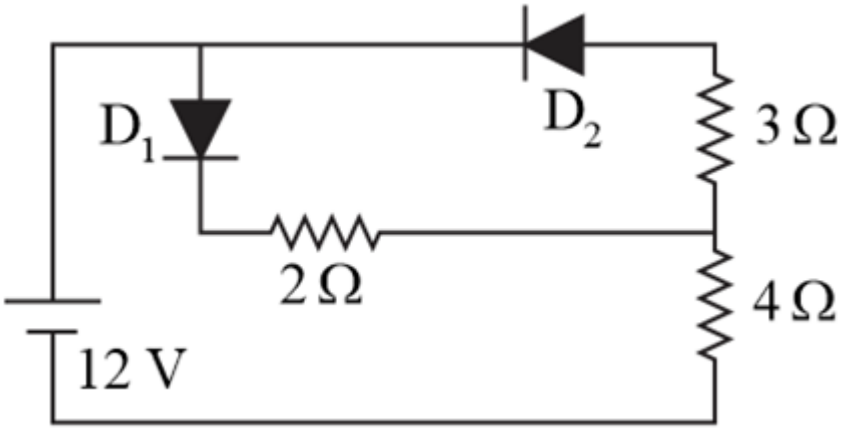
(2) If Assertion is true but Reason is false.

(3) If Assertion is false but Reason is true.

(4) If both Assertion and Reason are false.

Q10

In the circuit of the figure, treat diode as ideal, current in the 4 Ω resistor is



(1) 2 A

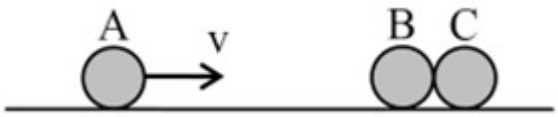
(2) 3 A

(3) $\frac{12}{7}$ A

(4) $\frac{30}{13}$ A

Q11

The figure below shows three identical balls A, B and C. Initially, the balls B and C are at rest and the ball A while moving with a velocity v, collides with ball B. If the collision is perfectly elastic, then after the collision



(1) all the three balls move with velocity $\frac{v}{2}$

(2) A comes to rest and $(B + C)$ moves with velocity $\frac{v}{\sqrt{2}}$

(3) A moves with velocity v and $(B + C)$ moves with velocity v

(4) A and B come to rest and C moves with velocity v

Q12

A wire of natural length l, Young's modulus Y and area of cross-section A is extended by x. Then the energy stored in the wire is given by

(1) $\frac{1}{2}\ \frac{YA}{l}x^2$

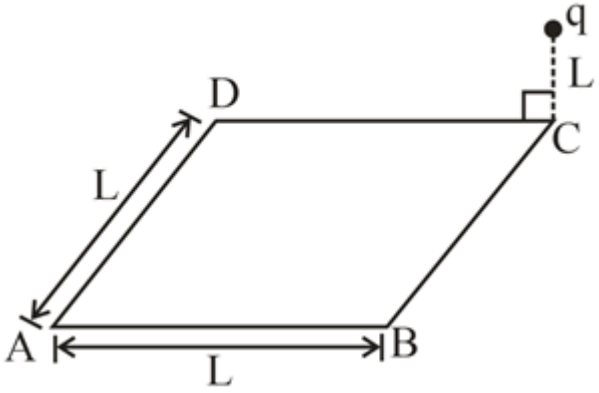
(2) $\frac{1}{3}\ \frac{YA}{l}x^2$

(3) $\frac{1}{2}\ \frac{Yl}{A}x^2$

(4) $\frac{1}{2}\ \frac{A}{Yl}x^2$

Q13

Point charge 'q' is placed above square face ABCD of side 'L' as shown in figure. Find flux related to the square.



(1) $\frac{q}{6\varepsilon_0}$

(2) $\frac{q}{12\varepsilon_0}$

(3) $\frac{q}{24\varepsilon_0}$

(4) zero

Q14

The density of a solid ball is to be determined in an experiment. The diameter of the ball is measured with a screw gauge, whose pitch is 0.5 mm and there are 50 divisions on the circular scale. The reading on the main scale is 2.5 mm and that on the circular scale is 20 divisions. If the measured mass of the ball has a relative error of 2%, the relative percentage error in the density is

(1) 0.9%

(2) 2.4%

(3) 3.1%

(4) 4.2%

Q15

The frequency and intensity of a light source are doubled. Consider the following statements and judge whether it is true of false-

1. Saturation photocurrent remains almost the same.

2. Maximum kinetic energy of the photoelectrons is doubled.

(1) Both I and II are true

(2) I is true but II is false.

(3) I is false but II is true.

(4) Both I and II are false.

Q16

A planet radiates heat at a rate proportional to the fourth power of its surface temperature T. If the steady temperature of the planet is due to an exactly equal amount of heat received from the sun then which of the following statements is true?

(1) The planet's surface temperature varies inversely as the distance of the sun

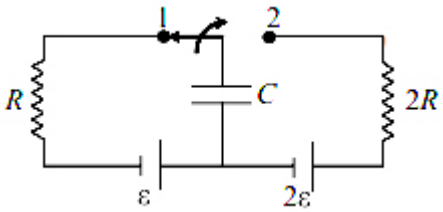
(2) The planet's surface temperature varies directly as the square of its distance from the sun

(3) The planet's surface temperature varies inversely as the square root of its distance from the sun

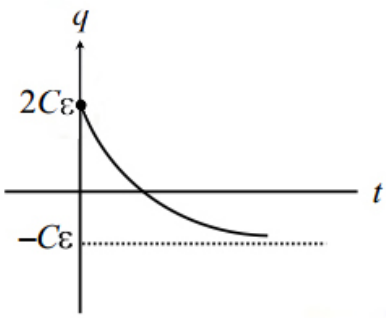
(4) The planet's surface temperature is proportional to the fourth power of distance from the sun

Q17

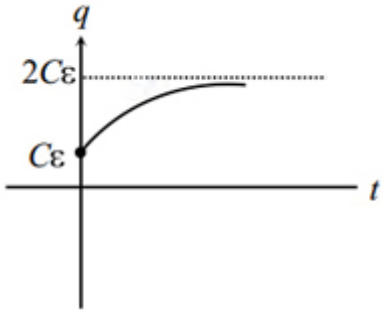
In the circuit shown in the figure, the switch is shifted from the position, $1 \rightarrow 2$ at $t = 0$. The switch was initially in position 1 for a long time. The graph between the charge on the upper plate of capacitor versus time t is



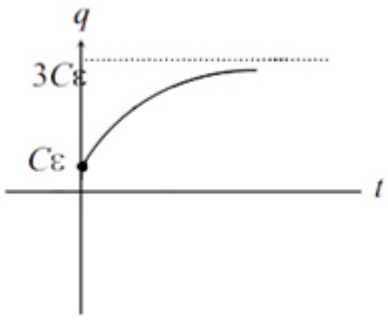
(1)



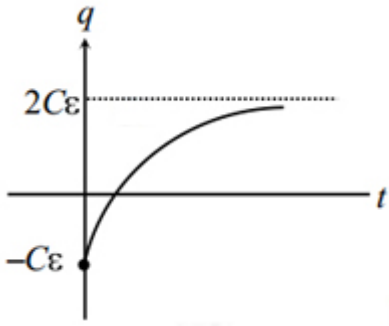
(2)



(3)



(4)



Q18

If pressure P, velocity V and time T are taken as fundamental physical quantities, the dimensional formula of force is

(1) PV^2T^2

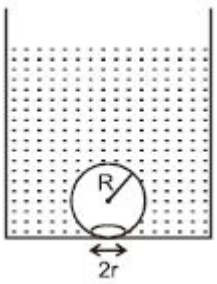
(2) $P^{-1}V^2T^{-2}$

(3) PVT^2

(4) $P^{-1}VT^2$

Q19

Bubbles beings formed at the bottom of the vessel detatch and rise on heating water. Take the bubles to be spheres of radius R and making a circular contact of radius r with the bottom of the vessel. If $r << R$, and the surface tension of water is T, value of r just before bubles detatch is: (density of water is ρ_w)



(1) $\mathrm{R}^2\sqrt{\frac{2\rho_{\mathrm{m}}\mathrm{g}}{3\mathrm{T}}}$

(2) $\mathrm{R}^2\sqrt{\frac{\rho_{\mathrm{a}}\mathrm{g}}{6\mathrm{T}}}$

(3) $\mathrm{R}^2\sqrt{\frac{\rho_{\mathrm{ag}}}{\mathrm{T}}}$

(4) $\mathrm{R}^2\sqrt{\frac{3\rho_{\mathrm{a}}\mathrm{g}}{\mathrm{T}}}$

Q20

Assertion: The drift velocity of electrons in a metallic wire will decrease, if the temperature of the wire is increased.

Reason: On increasing the temperature, conductivity of metallic wire decreases.

(1) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

(2) If both Assertion and Reason are true but Reason is not explanation of the Assertion.

(3) If Assertion is true but the Reason is false.

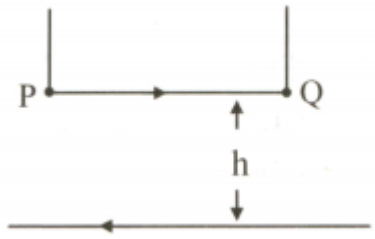
(4) If Assertion is false but Reason is true.

Q21

1 mole of monoatomic gas initially at $927^{\circ}\mathrm{C}$ undergoes a Carnot cycle. Each stage represents a compression or expansion in the ratio $1:8$. What is the lowest temperature of the engine? (in $^{\circ}\mathrm{C}$).

Q22

A long straight wire carrying current of $20\mathrm{A}$ rests on a table (as shown in figure). Another wire PQ of length $2\mathrm{m}$, mass $2.0\mathrm{g}$ carries the same current but in opposite direction. The wire PQ is such that it can slide freely up and down. The height upto which PQ rise is h, find 10h. (in cm) (Take $g = 10\mathrm{m\ s^{-2}}$)



Q23

In the Coolidge tube experiment, if the applied voltage is increased to three times, the short wavelength limit of continuous X-ray spectrum shifts by 20 pm. What is the initial voltage (in kV) applied to the tube to the nearest integer value? [Take $hc = 12400$ eV $\mathring{A}$, $e = 1.6 \times 10^{-19}$ C]

Q24

What is the peak voltage (in V) of the modulating signal that modulates the message signal wave of amplitude 20 V in order to have a modulation index of 75%?

Q25

H_2 gas undergoes a process described by the equation $PV^5 =$ constant. The amount given of H_2 gas is one mole.The heat capacity of the gas during this process is k times the value of gas constant (R). Value of $4k$ is _____.

Q26

In a compound microscope, the focal lengths of two lenses are 1.5 cm and 6.25 cm. An object is placed at 2 cm from objective so what is the distance (in cm) between two lenses if the final image is formed at 25 cm from eye lens?

Q27

The magnetic flux linked with a large circular coil, of radius R, is 0.5×10^{-3} Wb when a current of 0.5 A, flows through a small neighbouring coil of radius r, calculate the coefficient of mutual inductance for the given pair of coils (in mH).

Q28

A ball is thrown at some angle with the vertical such that its vertical component of the initial velocity is $20\ \mathrm{m\ s^{-1}}$. The wind is blowing horizontally such that it imparts $8\ \mathrm{m\ s^{-2}}$ horizontal acceleration on the ball. If the angle with vertical at which the ball must be thrown so that the ball returns to the man's hand is Φ, then what is $10\tan(\Phi) = ?$

Q29

In a resonance tube experiment, to determine the speed of sound in air, a pipe of diameter 5 cm is used. The air column in the pipe resonates with a tuning fork of frequency 480 Hz when the minimum length of the air column is 16 cm. Find the speed of sound (in $\mathrm{m\ s^{-1}}$) in the air at room temperature.

Q30

A sphere of radius R have volume charge density given as

$\rho(\mathrm{r}) = \mathrm{Kr}$ for $\mathrm{r} \leq \mathrm{R}$

$= 0$ for $\mathrm{r} > \mathrm{R}$

If electric field at distance $\frac{\mathrm{R}}{2}$ from centre is $\frac{\mathrm{KR}^2}{2\mathrm{N}\varepsilon_0}$ Then N will be.

Q31

This question contains two statements. This is Assertion-Reason type question. Statement - 1 (Assertion) and Statement - 2 (Reason). This question also has four alternative choices, only one of which is the correct answer. You have to select the correct choice.

Statement - 1: $\left[Fe(CN)_6\right]^{-4}$ ion has higher coagulation power than that of PO_4^{-3}, SO_4^{-2}, Br^- for

coagulation of positively charged sols.

Statement - 2: According to Hardy Schulze's rule the higher the valency of ions, the faster will be the precipitation or coagulation for the oppositely charged sol particles.

(1) Statement - 1 is True, Statement - 2 is True; Statement - 2 is a correct explanation for Statement - 1.

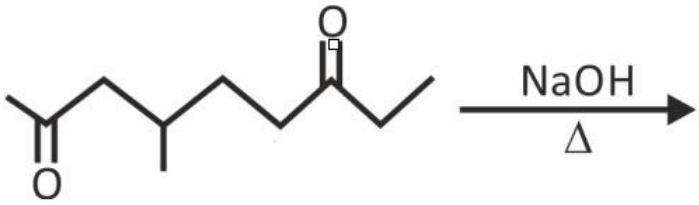
(2) Statement - 1 is True, Statement - 2 is True; Statement - 2 is not a correct explanation for Statement - 1.

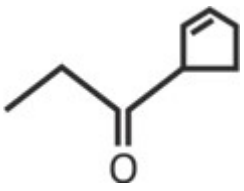
(3) Statement - 1 is True, Statement - 2 is False.

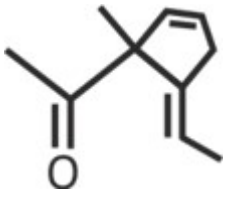
(4) Statement - 1 is False, Statement - 2 is True.

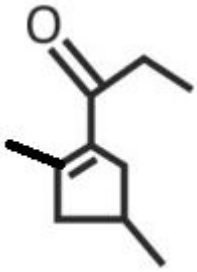
Q32

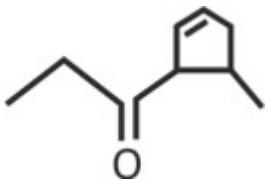
One of the product formed in the reaction is



(1) 

(2) 

(3) 

(4) 

Q33

The successive ionization energies of four elements are given as follows (in eV):

Elements	I. E_1	I. E_2	I. E_3	I. E_4
P	18.6	28.9	2068	4109
Q	12.8	1026	3125	5619
R	2486	3816	5139	8516
S	16.9	22.6	28.8	3214

Choose the correct statement from the following.

(1) The stable oxidation state of P is $+1$.

(2) The oxide of Q is acidic in nature.

(3) Element S forms electron deficient compounds.

(4) R is an alkaline earth metal.

Q34

Which process of purification is represented by the following series of reaction:

$$\underset{\text{(impure)}}{\text{Ti}} + 2\text{I}_2 \xrightarrow{250\ ^\circ\text{C}} \text{TiI}_4 \xrightarrow{1400\ ^\circ\text{C}} \underset{\text{(pure)}}{\text{Ti}} + 2\text{I}_2$$

(1) Cupellation

(2) Polling

(3) Van Arkel process

(4) Zone refining

Q35

The reagent used for softening the temporary hardness of water is :-

(1) $Ca_3(PO_4)_2$

(2) $CaCO_3$

(3) Na_2CO_3

(4) $NaOCl$

Q36

What is the minimum pH required to prevent the precipitation of ZnS in a solution that is 0.01 M $ZnCl_2$ and saturated with 0.10 M H_2S ?

[Given $K_{sp} = 10^{-21}$, $K_{a_1} \times K_{a_2} = 10^{-20}$]

(1) 0

(2) 1

(3) 2

(4) 4

Q37

Pick out incorrect statement.

(1) NF_3 molecule has trigonal pyramidal structure

(2) NF_3 is practically insoluble in water and is only hydrolyzed, when an electric spark is passed through a mixture with water vapour

(3) Dipole moment of NF_3 is more than that of NH_3

(4) Nitrogen (III) oxide (N_2O_3) is an acidic oxide

Q38

Match the geometry (given in column A) with the complexes (given in column B) in:

	Geometry: A		Complex: B
I	Octahedral	(P)	$[Ni(CN)_4]^{2-}$
II	Square planar	(Q)	$Ni(CO)_4$
III	Tetrahedral	(R)	$[Fe(CN)_6]^{4-}$

(1) $I-P, II-Q, III-R$

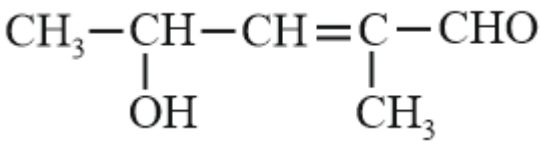
(2) $I-R, II-P, III-Q$

(3) $I-R, II-Q, III-P$

(4) $I-Q, II-P, III-R$

Q39

The IUPAC name of



(1) 4-Hydroxy-1-methylpentanal

(2) 4-Hydroxy-2-methylpent-2-en-1-al

(3) 2-Hydroxy-4-methylpent-3-en-5-al

(4) 2-Hydroxy-3-methylpent-2-en-5-al

Q40

Statement -1: The close packing of atoms in cubic structure is in the order FCC > BCC > SC because

Statement -2: Packing fraction: volume of the atom occupying unit cell /Vol. Of unit cell.

(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(2) Statement-1 is True, Statement-2 is True; Statement-2 NOT a correct explanation for Statement-1.

(3) Statement-1 is True, Statement-2 is False.

(4) Statement -1 is False, statement-2 is True.

Q41

Identify the pollutant gases largely responsible for the discoloured and lustreless nature of marble of the Taj Mahal.

(1) N_2 and CO_2

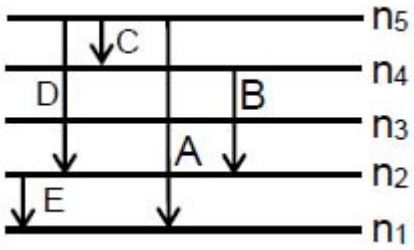
(2) SO_2 and O_3

(3) CO_2 and NO_2

(4) SO_2 and NO_2

Q42

Spectral lines for H-atom were observed as shown below where n_1 to n_5 are successive shells. If 'E' belongs to visible region, then which of the following statements are correct for the transition?



(A) There cannot be any line in UV region

(B) B and D lines belongs to Infrared region.

(C) Line having shortest wavelength is A

(D) Line having least energy is C

(1) (A), (B) and (C)

(2) (A) and (B)

(3) (C) and (D)

(4) (A), (B), (C) and (D)

Q43

Directions: Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

Assertion: Beryllium and aluminium chlorides both have Cl^- bridged chloride structure in solid phase.

Reason: Beryllium does not exhibit coordination number more than four as in valence shell there are only four orbitals.

(1) If both assertion and reason are true and reason is the correct explanation of assertion.

(2) If both assertion and reason are true but reason is not the correct explanation of assertion.

(3) If Assertion is true but reason is false.

(4) If both assertion and reason are false.

Q44

N_2 and O_2 are converted to monopositive cations N_2^+ and O_2^+ respectively. Which is incorrect?
(According to Molecular orbital theory)

(1) N_2^+ becomes diamagnetic

(2) In O_2^+ , paramagnetism decreases

(3) In O_2^+ , the bond order increases

(4) In N_2^+ the N-N bond is weakened.

Q45

The correct match between Item I and Item II is

Item I	Item II
(1) Norethindrone	(P) Anti-biotic
(2) Ofloxacin	(Q) Anti-fertility
(3) Equanil	(R) Hypertension
	(S) Analgesics

(1) 1—R; 2—P; 3—S

(2) 1—Q; 2—P; 3—R

(3) 1—R; 2—P; 3—R

(4) 1—Q; 2—R; 3—S

Q46

Which one of the following is NOT correct for monosaccharides?

1. They are optically active polyhydroxy carbonyl compounds

2. Fructose is ketose sugar and hence it does not give red precipitate with Fehling solution

3. α -D(+) glucose and β -D(+) glucose are anomers

4. Glucose and mannose are anomers

(1) 1, 2

(2) 2, 3

(3) 2, 4

(4) 1, 2, 3, 4

Q47

From which of the following tests, 1°, 2° and 3° amines can be distinguished?

(1) Action with HNO_2.

(2) Hinsberg reagent

(3) Acetylation

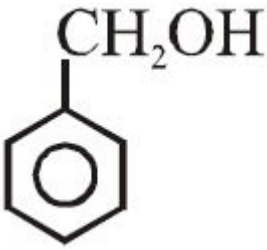
(4) Carbylamine reaction

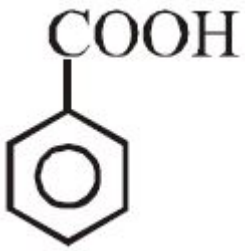
Q48

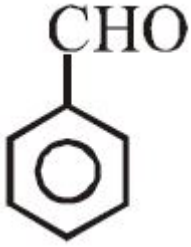
The product (C) in the following reaction is:

$$\text{Phenol} \xrightarrow[\Delta]{\text{Zn}} (\text{A}) \xrightarrow[\substack{\text{Anhydrous} \\ \text{AlCl}_3}]{\text{CH}_3\text{Cl}} (\text{B}) \xrightarrow[\Delta]{\text{KMnO}_4 / \text{OH}^{\ominus}} (\text{C}).$$

(1) 

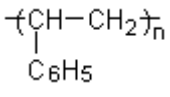
(2) 

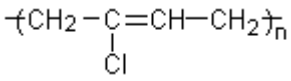
(3) 

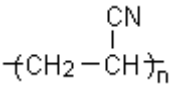
(4) 

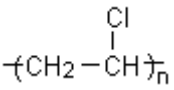
Q49

Which of the following structures represents Neoprene polymer?

(1) 

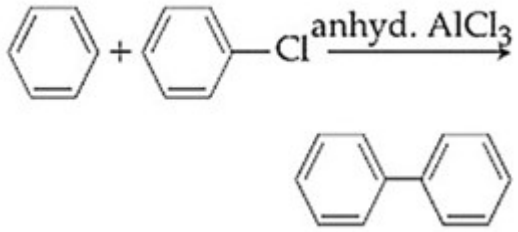
(2) 

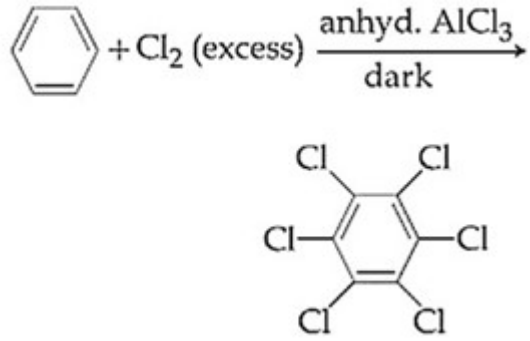
(3) 

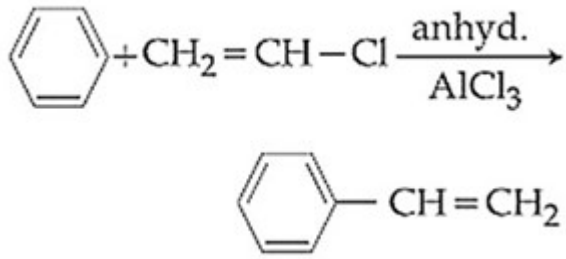
(4) 

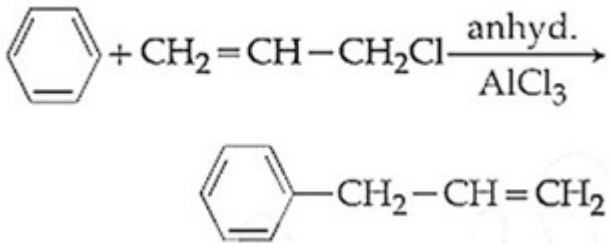
Q50

Consider the following reactions:

(a) 

(b) 

(c) 

(d) 

Which of these reactions are possible?

(1) (a) and (b)

(2) (a) and (d)

(3) (b), (c) and (d)

(4) (b) and (d)

Q51

As the temperature is raised from $20^{\circ}\mathrm{C}$ to $40^{\circ}\mathrm{C}$, the average kinetic energy of neon atoms changes by a factor of which of the following? (Multiply your answer with 293)

Q52

Copper sulphate solution $(250\ \mathrm{mL})$ was electrolysed using a platinum anode and a copper cathode. A constant current of $2\ \mathrm{mA}$ was passed for $16\ \mathrm{min}$. It was found that after electrolysis the absorbance of the solution was reduced to 50% of its original value. Concentration of copper sulphate in the solution to begin with is $\mathrm{x} \times 10^{-5}$. Find the value of x

Q53

Consider a first order gas phase decomposition reaction given below: $\mathrm{A}_{(g)} \longrightarrow \mathrm{B}_{(g)} + \mathrm{C}_{(g)}$

The initial pressure is 6.0 atm. The pressure drops to 3.0 atm after $6.93\ \mathrm{min}$. How much time (in minutes) would it take to lower the partial pressure of $\mathrm{A}_{(g)}$ by 4.0 atm? $[\text{Consider: } \log_{10}(3) = 0.48]$ (round off answer to nearest integer)

Q54

The oxidation state exhibited by the transition metal which is present in Ziegler-Natta catalyst is

Q55

Alcohol + Lucas reagent $\rightarrow$ Immediate turbidity

How many of the following will give above test positive?

$2-$Methylpropan$-2-$ol, butan$-1-$ol, $2-$methylpropan$-1-$ol, $2,2-$dimethylpropan$-1-$ol, $2,4-$dimethylpentan$-2-$ol, propan- $1,3-$diol

Q56

The total count of geometrical isomers for a square planar complex $\left[\mathrm{Mabcd}\right]^{\mathrm{n}\pm}$ is 'x'. The number of geometrical isomers possible for octahedral complex of the type $\left[\mathrm{Ma_4\,b_2}\right]^{\mathrm{n}\pm}$ is 'y'. Calculate x + y: _____ (a, b, c, d are monodentate ligands).

Q57

How many of the following solutions will behave as ideal solution, if the following substances are mixed together?

i. $\mathrm{n-Hexane\ +\ n-heptane}$

ii. $\mathrm{Bromoethane\ +\ chloroethane}$

iii. $\mathrm{Acetone\ +\ ethanol}$

iv. $\mathrm{Acetone\ +\ carbon\ \ disulphide}$

v. $\mathrm{Acetone\ +\ chloroform}$

vi. $\mathrm{Benzene\ +\ toluene}$

vii. $\mathrm{Phenol\ +\ aniline}$

Q58

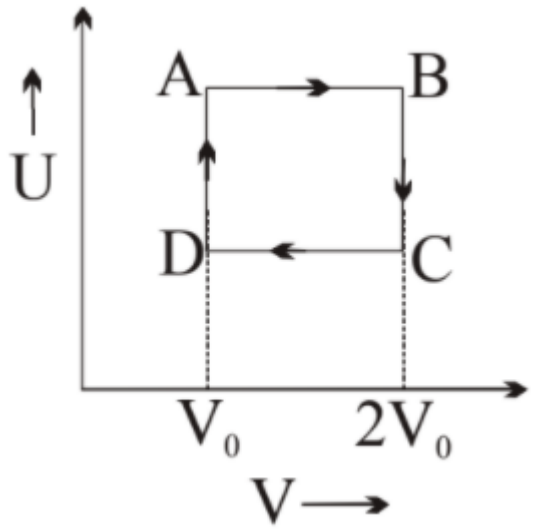
100 mL of 0.1M $KMnO_4$ solution reacts completely with 50 mL of $FeSO_4$ solution in acidic medium. The mass of $FeSO_4$ in 50 mL of $FeSO_4$ solution is M, find 10M. $\left[\text{ Molar mass of } FeSO_4 = 152 gmol^{-1}\right]$

Q59

For the reaction $3A \rightleftharpoons C + 2D$, initially A was taken. At equilibrium the concentration of D is twice that of A. The value of K_c is

Q60

The internal energy (U) of an ideal gas is plotted against volume for a cyclic process ABCDA, as shown in the figure.



The temperature of the gas at B and C are 500 K and 300 K, respectively. The heat absorbed by the gas (in cal / mol) in this cyclic process, is :

Q61

$\int_0^{10}[x]^3\{x\}dx$ is equal to:

[Note: Where [ ] and \{\} denotes greatest integer and fractional part functions respectively]

(1) 2025

(2) $\frac{2025}{2}$

(3) $\frac{2025}{4}$

(4) $\frac{2025}{8}$

Q62

The common tangent of two parabolas $y = x^2$ and $y = x^2 - 2x + 2$ is -

(1) $y = 2x - 1$

(2) $4y - x + 4 = 0$

(3) $4x - 4y - 1 = 0$

(4) $y + 4x + 4 = 0$

Q63

Let $f(x) = x^2 + \alpha x + \beta$ where $\alpha, \beta \in R$ and $f(f(x)) = 0$ has 2 roots 1 and 2 . Find the value of $2|f(0)|$.

(1) 3

(2) 6

(3) 4

(4) None of these

Q64

If the vector $a\bar{i} + \bar{j} + \bar{k}, \bar{i} + b\bar{j} + \bar{k}$ and $\bar{\mathrm{i}} + \bar{\mathrm{j}} + \mathrm{c}\bar{\mathrm{k}}, (\mathrm{a} \neq \mathrm{b} \neq \mathrm{c} \neq 1)$ are coplanar, then the value of

$\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} =$

(1) 1

(2) 2

(3) 3

(4) 4

Q65

The slope of normal at any point $(x, y), x > 0, y > 0$ on the curve $y = y(x)$ is given by $\frac{x^2}{xy - x^2y^2 - 1}$. If the curve passes through the point $(1, 1)$, then $e \cdot y(e)$ is equal to

(1) $\frac{1-\tan(1)}{1+\tan(1)}$

(2) $\tan(1)$

(3) 1

(4) $\frac{1+\tan(1)}{1-\tan(1)}$

Q66

If A and B are square matrices of order 3 such that $2(A + B) = A^T + B^T + 3I$ and $AA^T = 4I$, then find the value of det. $\left(12A^{-1} - BA^T + I\right)$

(1) 125

(2) 240

(3) 420

(4) 256

Q67

Let $S_n = {}^n\mathrm{C}_0 {}^n\mathrm{C}_1 + {}^n\mathrm{C}_1 {}^n\mathrm{C}_2 + \ldots\ldots\ldots + {}^n\mathrm{C}_{n-1} {}^n\mathrm{C}_n$ if $\frac{S_{n+1}}{S_n} = \frac{15}{4}$; find the sum of all possible values of $n (n \in N)$

(1) 1

(2) 6

(3) 4

(4) None of these

Q68

If $a_1, a_2, \ldots\ldots\ldots, a_{4001}$ are in arithmetic progression and $\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \ldots\ldots + \frac{1}{a_{4000} a_{4001}} = 10$ and $a_2 + a_{4000} = 50$. Find the value of $|a_1 - a_{4001}|$

(1) 15

(2) 24

(3) 30

(4) 42

Q69

Let $f(x) = x^3 - 3x^2 + 3x + 1$ and g be the inverse of f, then area bounded by the curve $y = g(x)$ and x-axis from $x = 1$ to $x = 2$, is -

(1) $\frac{1}{4}$ sq. units

(2) $\frac{7}{4}$ sq. units

(3) $\frac{9}{4}$ sq. units

(4) $\frac{3}{4}$ sq. units

Q70

If p and q are two statements, then which of the following statement is a tautology

(1) $p \Rightarrow (p \vee \sim q)$

(2) $(p \vee q) \Rightarrow p$

(3) $p \Rightarrow (p \wedge q)$

(4) $p \Leftrightarrow (p \Rightarrow q)$

Q71

If $\int \frac{\text{cosec}^2 x - 2019}{(\cos x)^{2019}} dx = \frac{-f(x)}{(g(x))^{2019}} + C$, then find the value of $\left| f\left(\frac{\pi}{4}\right) + g(0) \right|$. (where C is constant of integration.)

(1) 2

(2) 2.5

(3) 4

(4) 6

Q72

If $g(x)$ is twice differentiable real valued function satisfying $g''(x) - 3g'(x) > 3 \; \forall \; x \geq 0$ and $g'(0) = -1$, then $h(x) = g(x) + x \forall x > 0$ is :

(1) strictly increasing

(2) strictly decreasing

(3) non monotonic

(4) data insufficient

Q73

Let $A = \{1, 2, 3, 4\}$ and $B = \{0, 1, 2, 3, 4, 5\}$. The number of one-one functions from A to B which are not increasing is

(1) 320

(2) 345

(3) 360

(4) 375

Q74

In ΔPQR, the equation of the internal angle bisector of angle Q is $y = x$ and the equation of side PR is $x - 2y + 1 = 0$. If coordinates of P are $(3,2)$ and $2PQ = RQ$, then the coordinates of Q are

(1) $(3,3)$

(2) $(7,7)$

(3) $(-2, -2)$

(4) $(5,5)$

Q75

If $\mathrm{a}, \mathrm{b}, \mathrm{c}$ are non-zero real numbers, the system of equations $\mathrm{y} + \mathrm{z} = \mathrm{a} + 2\mathrm{x}, \; \mathrm{x} + \mathrm{z} = \mathrm{b} + 2\mathrm{y}$, $\mathrm{x} + \mathrm{y} = \mathrm{c} + 2\mathrm{z}$ is consistent and $\mathrm{b} = 4\mathrm{a} + \frac{\mathrm{c}}{4}$, then the sum of the roots of the equation $\mathrm{at}^2 + \mathrm{bt} + \mathrm{c} = 0$ is

(1) 3

(2) 2

(3) -2

(4) -3

Q76

If $f(x+y)=f(x)+f(y)-xy-1, \forall x, y \in R$ and $f(1)=1$, then the number of solutions of $f(n)=n, n \in N$, is

(1) one

(2) two

(3) no solution

(4) three

Q77

The sum of the infinite series $\cot^{-1}\left(\frac{7}{4}\right)+\cot^{-1}\left(\frac{19}{4}\right)+\cot^{-1}\left(\frac{39}{4}\right)+\cot^{-1}\left(\frac{67}{4}\right)+\ldots\ldots\infty$ is :

(1) $\frac{\pi}{4}-\cot^{-1}(3)$

(2) $\frac{\pi}{4}-\tan^{-1}(3)$

(3) $\frac{\pi}{4}+\cot^{-1}(3)$

(4) $\frac{\pi}{4}+\tan^{-1}(3)$

Q78

Let W denotes the words in the English dictionary. If the relation R is given by

$R=\{(x,\ y) \in W \times W:\ \text{the word}\ x \text{ and } y \text{ have at least one letter in common}\}$, then R is

(1) reflexive, symmetric and not transitive

(2) reflexive, symmetric and transitive

(3) reflexive, not symmetric and transitive

(4) not reflexive, symmetric and transitive

Q79

$L = \lim_{n\to\infty} \sqrt{n} \int_0^1 \frac{dx}{(1+x^2)^n}$

Suppose that the above limit exists, then choose the correct option.

(1) $\frac{1}{2} < L < 2$

(2) $4 < L < 5$

(3) $2 < L \leq 3$

(4) $L \geq 5$

Q80

The value of $\lim_{x\to 0} \frac{x\sin(\sin x) - \sin^2 x}{x^6}$ equals

(1) $\frac{1}{6}$

(2) $\frac{1}{12}$

(3) $\frac{1}{18}$

(4) $\frac{1}{24}$

Q81

The projection of the line $\frac{\mathrm{x}}{2} = \frac{\mathrm{y}-1}{2} = \frac{\mathrm{z}-1}{1}$ on a plane P is $\frac{\mathrm{x}}{1} = \frac{\mathrm{y}-1}{1} = \frac{\mathrm{z}-1}{-1}$. If plane P passes through $(a, -2, 0)$, then a is equal to

Q82

Let $S = S_1 \cap S_2 \cap S_3$ where :

- $S_1 = \{z | z \in C | z\ |, < 4\}$
- $S_2 = \left\{z \mid z \in C, I_m\left(\frac{z-1+i\sqrt{3}}{1-i\sqrt{3}}\right) > 0\right\}$
- $S_3 = \{z \mid z \in C, R(z) > 0\}$

If the area of S can be expressed as $\frac{a}{b}\pi$, where a and b are positive integers that are relatively prime. Find $a + b$

Q83

Let $\vec{a}=\begin{bmatrix}-3\\1\\1\end{bmatrix}, \vec{b}=\begin{bmatrix}4\\2\\4\end{bmatrix}$ & $\overrightarrow{c}=\begin{bmatrix}2\\2\\0\end{bmatrix}$. If V_1 is the volume of parallelopiped whose three coterminous edges are the vectors $\vec{a}+\vec{b}, \vec{b}+\vec{c}, \vec{c}+\vec{a}$ & V_2 is the volume of tetrahedron whose coterminous edges are the vectors $\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}$, then the value of (V_1+V_2) is

Q84

The mean and standard deviation of 50 observations are 50 and 10. Later on, it was decided to omit an observation that was incorrectly recorded as 99. If the variance of the remaining 49 observations is $\frac{\lambda}{49}$, then the value of λ is equal to

Q85

For all x, define the functions, $f(x)=x^2, g(x)=-x^2+1, p(x)=|x|+1$. Consider the piecewise function $h(x)$ where $h(x)=\begin{cases}\max(f(x),g(x),p(x)), & x\geq\frac{-81}{100}\\ \min(f(x),g(x),p(x)), & x<\frac{-81}{100}\end{cases}$.

If $h(x)$ is non-differentiable at m number of points and discontinuous at n number of points, then find the value of $(m+n)$.

Q86

A letter is known to have either from "TATA NAGAR" or from "CALCUTTA". On the envelope, just two consecutive letters TA are visible. If the probability that the letter came from 'TATA NAGAR" is in the form $\frac{p}{q}$, where p and q are co-prime positive integers, then find the value of $(p+q)$.

Q87

The number of quadratic polynomials $ax^2+2bx+c$ which satisfy the following conditions is k

(i) a, b, c are distinct

(ii) $a, b, c \in \{1,2,3,4, \ldots 2001,2002\}$

(iii) $x+1$ divides $ax^2+2bx+c$.

Then $\frac{k}{10^3}$ is equal to

Q88

Let $A=\begin{bmatrix} a & x & p \\ y & q & b \\ r & c & z \end{bmatrix}$ and $B=\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ where $a, b, c, x, y, z, p, q, r$ are natural numbers.

If tr. $\left(AB+AB^3+AB^5+\ldots+AB^{19}\right)=210$, then find number of ordered triplets (p, q, r).

[Note: tr. (P) denotes the trace of matrix P.]

Q89

Find the sum of all integral values of a for which all the roots of the equation $x^4-4x^3-8x^2+a=0$ are real.

Q90

The vertices of the triangle ABC are $A(0,0),\ B(3,0)$ and $C(3,4)$, where A and C are foci of an ellipse and B lies on the ellipse. If the length of the latus rectum of the ellipse is $\frac{12}{p}$ units, then the value of 2p is

Answer Key

Q1 (1)	**Q2** (3)	**Q3** (3)	**Q4** (4)
Q5 (4)	**Q6** (2)	**Q7** (1)	**Q8** (3)
Q9 (1)	**Q10** (1)	**Q11** (4)	**Q12** (1)
Q13 (3)	**Q14** (3)	**Q15** (4)	**Q16** (3)
Q17 (4)	**Q18** (1)	**Q19** (1)	**Q20** (2)
Q21 (27)	**Q22** (8)	**Q23** (41)	**Q24** (15)
Q25 (9)	**Q26** (11)	**Q27** (1)	**Q28** (8)
Q29 (336)	**Q30** (8)	**Q31** (1)	**Q32** (3)
Q33 (3)	**Q34** (3)	**Q35** (3)	**Q36** (2)
Q37 (3)	**Q38** (2)	**Q39** (2)	**Q40** (1)
Q41 (4)	**Q42** (4)	**Q43** (2)	**Q44** (1)
Q45 (2)	**Q46** (3)	**Q47** (2)	**Q48** (3)
Q49 (2)	**Q50** (4)	**Q51** (313)	**Q52** (8)
Q53 (11)	**Q54** (4)	**Q55** (2)	**Q56** (5)
Q57 (3)	**Q58** (76)	**Q59** (4)	**Q60** (280)
Q61 (2)	**Q62** (3)	**Q63** (1)	**Q64** (1)
Q65 (4)	**Q66** (1)	**Q67** (2)	**Q68** (3)

Q69 (1)	**Q70** (1)	**Q71** (1)	**Q72** (1)
Q73 (2)	**Q74** (2)	**Q75** (4)	**Q76** (1)
Q77 (3)	**Q78** (1)	**Q79** (1)	**Q80** (3)
Q81 (5)	**Q82** (23)	**Q83** (288)	**Q84** (2550)
Q85 (4)	**Q86** (18)	**Q87** (2002)	**Q88** (190)
Q89 (6)	**Q90** (7)		

Q1

$\mathrm{U(x)=\left(x^2-3x\right)J}$

For a conservative field, Force, $\mathrm{F=-\frac{dU}{dx}}$

$\therefore \quad \mathrm{F=-\frac{d}{dx}\left(x^2-3x\right)=-(2x-3)=-2x+3}$

At equilibrium position, $\mathrm{F=0}$

$-2x+3=0 \quad \Rightarrow \quad \mathrm{x=\frac{3}{2}m=1\cdot 5\ m}$

Q2

$\mathrm{T \propto R^{\frac{3}{2}}}$

$\frac{(T+\Delta T)}{T}=\left(\frac{R+\Delta R}{R}\right)^{3/2}$

$1+\frac{\Delta T}{T}=\left(1+\frac{\Delta R}{R}\right)^{3/2}$

$\therefore \Delta R << R$

$1+\frac{\Delta T}{T}=1+\frac{3}{2}\frac{\Delta R}{R}$

$\frac{\Delta T}{T}=\frac{3}{2}\frac{\Delta R}{R}$

$\Delta T=\frac{3}{2}T\cdot\frac{\Delta R}{R}$

Q3

The bridge will be balanced when the shunted resistance of the value of $3\ \Omega$

$\therefore 3=\frac{4\times R}{4+R}$

$\Rightarrow 12+3R=4R$

$\Rightarrow R=12\ \Omega$

Q4

Each half part has a focal length f. The rays passing through first part will converge downwards while passing through the second part, the rays will equally bend upwards. So the power of the combination will be zero.

Q5

Total MI of the system,

$\mathrm{I=I_1+I_2+I_3}$

$I_2 = I_3 = \frac{2}{3}mr^2 + mr^2 = \frac{5mr^2}{3}$

$I_1 = \frac{2}{3}mr^2$

$\therefore I = 2 \times 5\frac{mr^2}{3} + \frac{2}{3}mr^2$

$= \frac{12mr^2}{3} = 4\ mr^2$

Q6

Path difference on circles around 'P' is same. So, The fringes obtained on the screen in the given condition will be concentric circles.

Q7

After first half hour with the process A ,

$N = \frac{N_0}{2}$

for next one hour with the process B , i.e. $t = \frac{1}{2}$ hr to $t = 1\frac{1}{2}$ hr

$N = \frac{N_0}{2}\left(\frac{1}{2}\right)^4 = N_0\left(\frac{1}{2}\right)^5$

Half life for both A and B ,

$\frac{1}{t_{1/2}} = \frac{1}{1/2} = \frac{1}{1/4}$

$\frac{1}{t_{1/2}} = 2 + 4$

$t_{1/2} = \frac{1}{6}hr$

for further half an hour with both A and B , i.e. $t = 1\frac{1}{2}$ hr to $t = 2$ hr

$N = \left[N_0\left(\frac{1}{2}\right)^5\right]\ \left[\frac{1}{2}\right]^3 = N_0\left[\frac{1}{2}\right]^8$

Q8

Let the rod be depressed by a small amount x (in the figure). Both the springs are compressed by x. When the rod is released, the restoring torque is given by

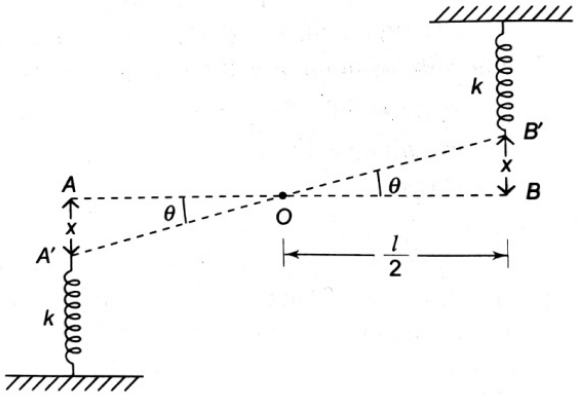
$\tau = (kx) \times \frac{l}{2} + (kx) \times \frac{l}{2} = (kx)l$

Now $\tan\theta = \frac{x}{l/2} = \frac{2x}{l}$.

Since θ is small, $\tan\theta \simeq \theta$, where θ is expressed in radian.

Thus $\theta = \frac{2x}{l}$ or $x = \frac{\theta l}{2}$

$\tau = k\left(\frac{\theta l}{2}\right) \times l = \frac{k\theta l^2}{2}$



If I is the moment of inertia of the rod about O, then

$I\frac{d^2\theta}{dt^2} = -\left(\frac{kl^2}{2}\right)\theta$

Or $\frac{d^2\theta}{dt^2} = -\left(\frac{kl^2}{2I}\right)\theta$

Since $\frac{d^2\theta}{dt^2} \propto (-\theta)$, the motion is simple harmonic whose angular frequency is given by

$\omega = \sqrt{\frac{kl^2}{2I}}$

Now, $\omega = \frac{2\pi}{T}$ and $I = \frac{ml^2}{12}$. Therefore, we have

$\frac{2\pi}{T} = \sqrt{\frac{kl^2}{2} \times \frac{12}{ml^2}} = \sqrt{\frac{6k}{m}}$

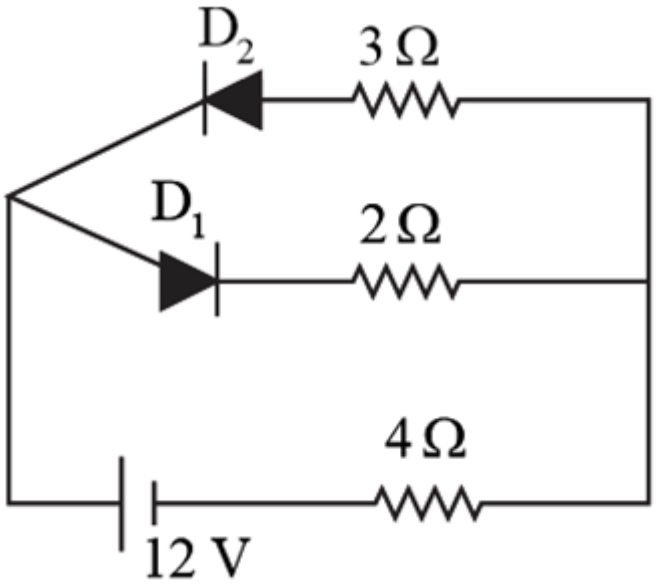
Or $T = \pi\sqrt{\frac{2m}{3k}}$, which is choice $\pi\sqrt{\frac{2m}{3k}}$

Q9

According to Kepler's law, all the planets move in elliptical orbits with the sun as one of the foci. The comets are frozen leftovers from the formation of the solar system composed of dust, rock and ices. Comets do not revolve in an elliptical orbit around the sun, so they do not obey Kepler's law.

Q10

The given circuit can be redrawn as,



This means, D_2 is reversed based. Hence, this arm of the circuit will act as open arm i.e., the close circuit will have two resistors 2 Ω and 4 Ω which will be in series connection.

$\Rightarrow \ R_{eq} = 2\ \Omega + 4\ \Omega$

$= 6\ \Omega$

So, current through 4 Ω resistor will be,

$I = \frac{V}{R}$

$= \frac{12}{6}$

$= 2\ \text{A}$

Q11

If two identical balls undergo elastic head-on collision then their velocities are exchanged.

Therefore in the collision of A and B, A comes to rest and B acquires velocity v.

Subsequently, in the collision of B and C, B comes to rest and C acquires the velocity v.

Q12

Energy stored in the wire

$U = \frac{1}{2}Y \times \left(\text{strain}\right)^2 \times \text{volume}$

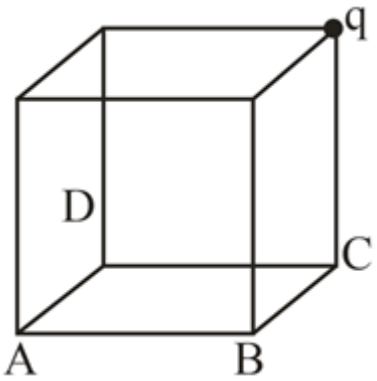
or $U = \frac{1}{2}Y \times \left(\frac{x}{l}\right)^2 \times Al$

or $U = \frac{1}{2}\frac{Yx^2}{l} \times A$

or $U = \frac{1}{2}\frac{YA}{l}x^2$

Q13

Complete the square by symmetric faces



So charge will appear at corner of the cube

So $\phi_{\text{ABCD}} = \frac{\text{q}}{24\varepsilon_0}$

Q14

Least count $= \frac{0.5}{50} = 0\ .01\ \ \text{mm}$

Diameter of ball $D = 2.5\ \text{mm} + (20)(0\ .01)$

$D = 2.7\ \text{mm}$

$\rho = \frac{M}{vol} = \frac{M}{\frac{4}{3}\pi\left(\frac{D}{2}\right)^3}$

$\left(\frac{\Delta\rho}{\rho}\right)_{max} = \frac{\Delta M}{M} + 3\frac{\Delta D}{D};\ \left(\frac{\Delta\rho}{\rho}\right)_{max} = 2\% + 3\left(\frac{0.01}{2.7}\right) \times 100\%$

$\frac{\Delta\rho}{\rho} = 3.1\%$

Q15

The saturation photocurrent (i) depends on intensity (I) of light ie,

$i \propto I$.

So, when intensity changes, the saturation current also changes. Hence the statement I false.

$hv = \phi + KE_{\max}$

when v is doubled KE will change but it will be not be doubled.

Q16

Rate of loss of energy by unit area of the planet $= \sigma \mathrm{T}^4$, where σ is the Stefan's constant. Let Q be the total energy emitted by the sun every second. If d is the distance of the planet from sun, then Q falls uniformly over the inner surface of the sphere of radius d. Rate of gain of heat by unit area of planet,

$$= \frac{\mathrm{Q}}{4\pi \mathrm{d}^2}$$

For steady temperature of planet,

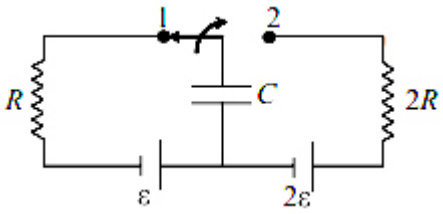
$$\sigma \mathrm{T}^4 = \frac{\mathrm{Q}}{4\pi \mathrm{d}^2}$$

$$\mathrm{T}^4 = \frac{\mathrm{Q}}{4\pi\sigma \mathrm{d}^2} \text{ or } \mathrm{T} = \left(\frac{\mathrm{Q}}{4\pi\sigma \mathrm{d}^2}\right)^{1/4}$$

$$\mathrm{T} \propto \frac{1}{\sqrt{\mathrm{d}}}$$

Q17

In the given circuit when the switch is in position 1.



Then the charge variation with time is

$$q = c\varepsilon\left(e^{-\frac{t}{Rc}} - 1\right)$$

thus, at $t = \infty,\ q = -c\varepsilon$

Now when the switch is changed to position 2, then the charge variation with time will be given by

$$q = c\varepsilon\left(2 - 3e^{\frac{-t}{2Rc}}\right)$$

$\therefore$At $t = 0,\ q = -c\varepsilon$

and at $t = \infty,\ q = 2c\varepsilon$

Q18

Let $F \propto P^x V^y T^z$

by substituting the following dimensions :

$\left[P\right] = \left[\mathrm{ML^{-1}\,T^{-2}}\right],\ \left[V\right] = \left[\mathrm{LT^{-1}}\right],\ \left[T\right] = \left[\mathrm{T}\right]$

and comparing the dimension of both sides

$x = 1,\ y = 2,\ z = 2$

$F = PV^2T^2$

Q19

On heating water

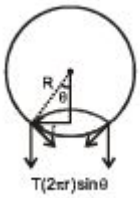
$s\sin\theta = \frac{\mathrm{r}}{\mathrm{R}}$

The bubble will detach if Buoyant force 2 Surface tension force

$(\rho_{\mathrm{w}})\left(\frac{4}{3}\pi\mathrm{R}^3\right)\mathrm{g} \geq \left(\mathrm{T}\right)\left(2\pi\mathrm{r}\right)\sin\theta$

Solving $\mathrm{r} = \sqrt{\frac{2\mathrm{p_w R^4 g}}{3\mathrm{T}}}$

$\sin\theta = \frac{\mathrm{r}}{\mathrm{R}}$



Q20

On increasing temperature, KE of free electron increases. They collide more rapidly hence drift velocity decreases. Also with increase in temperature resistance increases, so conductivity decreases.

∴ Both statements are true, but reason is not correct explanation of assertion.

Q21

In Carnot cycle, temperature varies during adiabatic change. For adiabatic change,

$\mathrm{T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}} \ldots . \left(\mathrm{i}\right)$

Given that, $\mathrm{V_1 : V_2 = 1 : 8}$,

$\gamma = \frac{5}{3}$(for monoatomic gas)

$T_1 = 927 + 273$

$= 1200\ \mathrm{K}$

$\mathrm{T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1}} \quad \ldots . . [\text{From } (i)]$

$= 1200 \times \left(\frac{1}{8}\right)^{\frac{5}{3}-1}$

$= 1200 \times \left(\frac{1}{8}\right)^{\frac{2}{3}}$

$= 1200 \times \frac{1}{4}$

$= 300\mathrm{K}$

$\therefore \ \mathrm{T_2} = 300 - 273 = 27^{\circ}\mathrm{C}$

Q22

The magnetic field produced by long-straight wire carrying current of $20\mathrm{A}$ rests on a table on small wire,

$\mathrm{B} = \frac{\mu_0 \mathrm{I}}{2\pi \mathrm{h}}$ .... (i)

The magnetic force on smal conductor is

$\mathrm{F_B = I/\,Bsin\,\theta = I/B = Il\left(\frac{\mu_0 I}{2\pi h}\right)}$

$= \frac{\mu_0 I^2 l}{2\pi h} \quad \ldots . \left(\text{ii} \right)$

At equilibrium,

$\Rightarrow \mathrm{F_B} = \mathrm{F_G}$

$\Rightarrow \frac{\mu_0 I^2 l}{2\pi h} = mg$

$\Rightarrow \mathrm{h} = \frac{\mu_0 \mathrm{I}^2 l}{2\pi\,\mathrm{mg}} = \frac{4\pi \times 10^{-7} \times 20 \times 20 \times 2}{2\pi \times 2 \times 10^{-3} \times 10}$

$= 8 \times 10^{-3} \mathrm{m}$

$\therefore \quad \mathrm{h} = 0.80\,\mathrm{cm}$

Q23

$\because$ the cut-off wavelength for continuous x-ray is given as : $\lambda_{\min} = \frac{hc}{eV_0}$

$\therefore$

$\frac{hc}{eV_0} = \lambda \ldots \; Eq\left(1.\right)$

$\frac{hc}{3eV_0} = \lambda - \Delta\lambda \ldots \; Eq\left(2.\right)$

$(\mathrm{V_0} = \text{accelerating voltage})$

$\Rightarrow \quad V_0 = \frac{2hc}{3e\Delta\lambda} = \frac{2 \times 6.6 \times 10^{-34} \times 3 \times 10^8}{3 \times 1.6 \times 10^{-19} \times 20 \times 10^{-12}} \approx 41\ kV$

Q24

$\mathrm{As}, \mathrm{m_a} = \frac{\mathrm{A_m}}{\mathrm{A_c}}$

$\therefore \quad A_m = m_a \times A_c = 0.75 \times 20 = 15\mathrm{V}$

Q25

Given, PV^5 =constant

This is polytropic process with $n = 5$.

∴Specific heat capcity,

$C = C_V + \frac{R}{1-n}$

$= \frac{R}{\gamma-1} + \frac{R}{1-n}$

$= \frac{\mathrm{R}}{\frac{7}{5}-1} + \frac{\mathrm{R}}{1-5} \quad \ldots\ldots \left(\because \gamma_{\mathrm{di}} = \frac{7}{5}\right)$

$= \frac{5R}{2} - \frac{R}{4} = \frac{9R}{4}$

$\therefore \quad C = (2.25)R$

Q26

From lens maker's equation,

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

For objective,

$\frac{1}{f_o} = \frac{1}{v_o} - \frac{1}{u_o}$

$\therefore \quad v_0 = \frac{u_0 f_0}{(u_0 + f_0)} \qquad \ldots (i)$

For eye-piece,

$\frac{1}{\mathrm{f_e}} = \frac{1}{\mathrm{V_e}} - \frac{1}{\mathrm{u_e}}$

Here,$v_e = D = 25\,\mathrm{cm}$

$\therefore \quad \frac{1}{\mathrm{u_e}} = \frac{1}{\mathrm{D}} - \frac{1}{\mathrm{f_c}}$

$\therefore \quad \mathrm{u_e} = \frac{\mathrm{f_e D}}{\mathrm{f_e - D}} \quad \ldots (ii)$

From equations (i) and (ii),

$\mathrm{v_0} = \frac{(-2)(1.5)}{(-2)+(1.5)} = 6$

$\mathrm{u_e} = \frac{(6.25)(-25)}{(6.25)-(-25)} = -5$

As negative sign denotes direction neglecting it,

$\therefore \quad L = v_0 + u_\mathrm{e} = 6 + 5 = 11\,\mathrm{cm}$

Q27

Given,

Magnetic flux linked with the larger coil of radius R is $\phi = 0.5 \times 10^{-3}$ Wb

Strength of current through small neighbouring coil $I = 0.5$ A

The total amount of magnetic flux linked with all the turns of one coil is directly proportional to the strength of current in neighbouring coil.

$\phi = MI$

$\Rightarrow M = \frac{\phi}{I}$

Substituting the values, we get

$M = \frac{0.5 \times 10^{-3}}{0.5} = 1 \text{ mH}$

So, the coefficient of mutual inductance for the given pair of coils is 1 mH.

Q28

In y direction

$u_y = 20 m/s$

$a_y = -10 m/s^2$

$s_y = u_y t + \frac{1}{2} a_y t^2$

$t = \frac{2u_y}{a_y} = \frac{2 \times 20}{10} = 4sec$

In - α direction

$s_x = 0 = u_x t + \frac{1}{2} a \times d^2$

$0 = u_x(4) - \frac{8}{2}(4)^2$

$16 = 4x$

$\tan\theta = \frac{4x}{4y} = \frac{16}{20}$

$10\tan\theta = 10 \times \frac{16}{20} = 8$

Q29

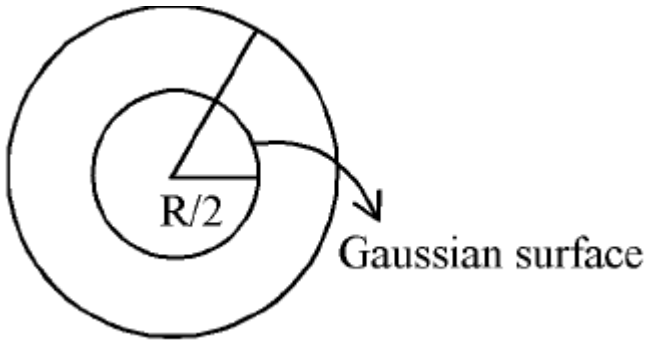
$\frac{v}{4[L+0.6r]} = 480$ (frequency)

$\Rightarrow v = 480 \times 4 \times [L + 0.6r]$

$\Rightarrow v = 480 \times 4 \times [0.16 + 0.6 \times 0.025]$

$\Rightarrow v = 336 \text{ m s}^{-1}$

Q30



$\oint \vec{\mathrm{E}} \cdot \mathrm{d}\vec{\mathrm{s}} = \frac{\mathrm{q_{in}}}{\varepsilon_0}$

$\mathrm{E}4\pi\left(\frac{\mathrm{R}}{2}\right)^2 = \frac{\int_0^{\mathrm{R}/2} \rho 4\pi \mathrm{r}^2 \, \mathrm{dr}}{\varepsilon_0}$

$$\Rightarrow \quad E\pi \mathrm{R}^2 = \frac{\int_0^{\mathrm{R}/2} \mathrm{Kr}\, 4\pi \mathrm{r}^2\, \mathrm{dr}}{\varepsilon_0}$$

$$\Rightarrow \quad E = \frac{4\mathrm{K}\int_0^{\mathrm{R}/2} \mathrm{r}^3\, \mathrm{dr}}{\mathrm{R}^2\varepsilon_0}$$

$$\Rightarrow \quad E = \frac{4\mathrm{K}}{\mathrm{R}^2\varepsilon_0}\left[\frac{r^4}{4}\right]_0^{R/2}$$

$$\Rightarrow \quad E = \frac{\mathrm{K}}{\mathrm{R}^2\varepsilon_0}\left[\frac{R^4}{16}\right]$$

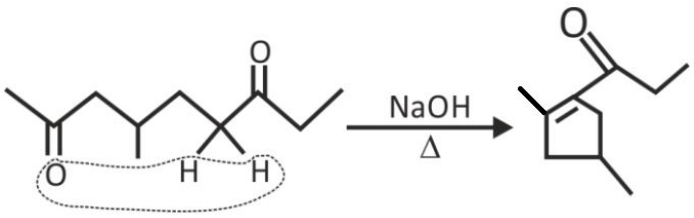
$$\Rightarrow \quad E = \frac{\mathrm{KR}^2}{2\times 8\varepsilon_0}$$

Q31

Coagulation power is directly proportional to charge on the ions.

Statement - 1 is True, Statement - 2 is True; Statement - 2 is a correct explanation for Statement - 1.

Q32



Q33

Element ‘S’ forms electron deficient compounds.

Q34

Van Arkel's method is used to purify crude titanium metal. It is heated with iodine to about 500K to form volatile compound. TiI_4 leaving behind the impurities . TiI_4 is further heated to 1700K when it decomposes to give pure titanium.

Q35

The reason for temporary hardness is owing to the presence of bicarbonates of Ca and Mg.

To remove this temporary hardness, we have to use agents that can precipitate out the bicarbonates in an easy-to-filter-out form:

$\mathrm{Ca(HCO_3)_2 + Na_2CO_3 \rightarrow CaCO_3 \downarrow + NaHCO_3}$

$\mathrm{Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 \downarrow + 2H_2O}$

The hypochlorite anion is oxidizing and gets hydrolyzed to $^{-}\mathrm{OH}$.

$\mathrm{OCl^{-}(aq) + H_2O(I) \rightleftharpoons HOCl(aq) + {}^{-}OH(aq)}$

Calcium phosphate is just insoluble in water. It is useful in this context.

Q36

$\mathrm{K_{sp}\ for\ ZnS = \left[Zn^{2+}\right]\left[S^{2-}\right]}$

$\left[\mathrm{S^{2-}}\right] = \frac{10^{-21}}{0.01} = 10^{-19}$

$\mathrm{H_2S \rightleftharpoons H^{+} + HS^{-}}$

$\mathrm{K_{a_1} = \frac{\left[H^{+}\right]\left[S^{2-}\right]}{\left[HS^{-}\right]}}$

$\mathrm{HS^{-} \rightleftharpoons H^{+} + S^{2-}}$

$\mathrm{K_{a_2} = \frac{\left[H^{+}\right]\left[S^{2-}\right]}{\left[HS^{-}\right]}}$

for $\mathrm{K_{a_1}.\ K_{a_2} = \frac{\left[H^{+}\right]^2\left[S^{2-}\right]}{\left[H_2S\right]}}$

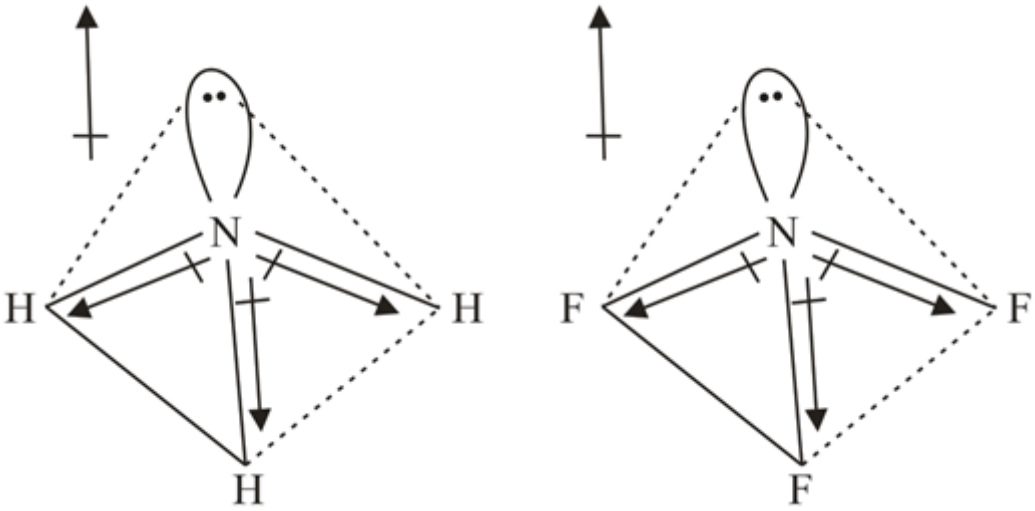
$10^{-20} = \frac{\left[\mathrm{H^{+}}\right]^2 \times 10^{-19}}{0.1} \Rightarrow \left[\mathrm{H^{+}}\right] = 0.1$

or $\mathrm{pH} = 1$

Q37

Dipole moment of $\mathrm{NH_3}$ (0.49×10^{-29} cm) is more than that of $\mathrm{NF_3}\left(0.07 \times 10^{-29}\mathrm{cm}\right)$. This is due to the fact that the direction of the dipole moments of the bonding $\mathrm{N-H}$ and non-bonding electron pairs coincide in $\mathrm{NH_3}$ and therefore, vector addition of all dipole moments yield a large resultant dipole moment. In $\mathrm{NF_3}$, on

the contrary, the dipole moments of $N-F$ bonds and of the non-bonding electron pair are in opposite directions, so that they are partially compensated when summated.



A. From the above figre, it is clear that NF_3 has a trigonal pyramidal structure.

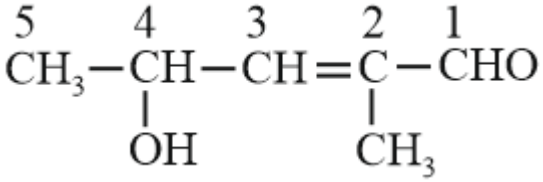
B. $2NF_3 + 3H_2O(\text{vapour}) \rightarrow N_2O_3 + 6HF$

D. $N_2O_3(l) + H_2O(l) \rightarrow \underset{(\text{Acid})}{2HNO_2(l)}$

Q38

In $\left[Fe(CN)_6\right]^{4-}$ Fe is in $+2$ state with elctronic configuration $4s^o3d^6$ and CN^- is strong ligand causes inward pairing, thus shows d^2sp^3 , Octahedral. $Ni(CO)_4$ is sp^3 tetrahedral and $\left[Ni(CN)_4\right]^{2-}$ is dsp^2 , square planar.

Q39



4-Hydroxy-2-methylpent-2-en-1-al

Q40

An FCC contains 4 atoms per unit cell whereas BCC and SC contain 2 and 1 atoms respectively.

The Packing fraction for FCC= 0.74

The Packing fraction for BCC= 0.68

The Packing fraction for SC= 0.52

Q41

Industries present nearby Taj Mahal produce a lot of NO_2 and SO_2 gases which react with water, oxygen and other chemicals to form sulphuric acid and nitric acid. These then mix with water and make the rain acidic which than react with marble to decolourise it as follows

$CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + CO_2 + H_2O$

Q42

For H-atom E belongs to visible region when $n_1 = 2$and $n_2 = 3$. B, C and D lines belongs to Infrared region.

Q43

$BeCl_2$ and $AlCl_3$ both have bridged structure in solid phase.

Beryllium does not exhibit coordination number more than four as in valence shells there are only four orbitals. The remaining members of the group can have a coordination number of six by making use of d-orbitals.

Q44

M.O. configuration of N_2 is

$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi^2 {p_x}^2 = \pi 2{p_y}^2 < \sigma 2{p_z}^2$

B.O. of $N_2 = \frac{10-4}{2} = 3$

M.O. configuration of N_2^+ is

$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2{p_x}^2 = \pi 2{p_y}^2 < \sigma 2{p_z}^1$

B.O. of $N_2^+ = \frac{9-4}{2} = 2.5$

M.O. of configuration of O_2 is

$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2p_z{}^2 < \pi 2p_x{}^2 < \pi 2p_y{}^2 < \pi^* 2p_x{}^1 < \pi^* 2p_y{}^1$

$\therefore$ B.O. of $O_2 = \frac{10-6}{2} = 2$

As it is paramagnetic so option (a) is incorrect.

M.O. configuration of O_2^+ is

$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2p_x{}^2 < \pi 2p_y{}^2 \cong \pi 2p_z{}^2 < \pi^* 2p_y{}^1 \cong \pi^* 2p_z{}^0$

B.O. of $O_2^+ = \frac{10-5}{2} = 2.5$

B.O. $\propto$ Bond energy $\propto \frac{1}{\text{Bond length}}$

Q45

(1) Norethindrone — Antifertility drug

(2) Ofloxacin — Anti – Biotic

(3) Equanil — Hypertension (tranquailizer)

Q46

Statement

1. Correct

2. Incorrect- Including fructose all monosaccharides give the fehling's test

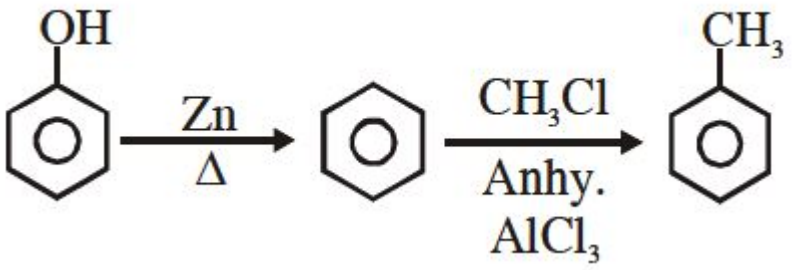
3. Correct

4. Incorrect- Glucose & mannose are epimers not anomers.

Q47

Mixture of $1^\circ, 2^\circ$ and 3° amines can be distinguished by Hinsberg's method. In Hinsberg's method benzene sulphonyl chlotide $C_6H_5SO_2Cl$ (Hinsberg's reagent) is used. 1° and 2° amines reacts to give sulphonamide derivatives with loss of HCl, whereas 3° amines do not give any isolable products other than the starting amine. In the latter case a quanternary ammonium salt may be as an intermediate, but this rapidly breaks down in water to liberate the original 3° amine.

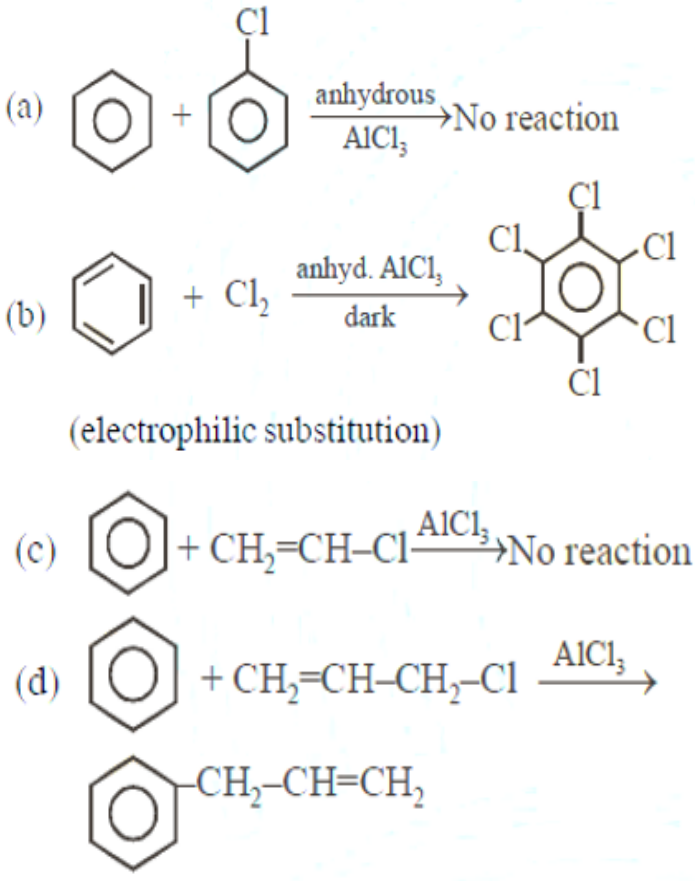
Q48



Q49

As neoprene is a polymer of 2- chloro buta 1, 3- di-ene $-(CH_2-\underset{\underset{Cl}{|}}{C}=CH-CH_2-)_n$

Q50



Q51

Average kinetic energy $\overline{\mathrm{KE}}$ (per molecule)$= \frac{3}{2}\mathrm{k_b T}$

$\frac{\overline{\mathrm{KE}}(40^{\circ}\mathrm{C})}{\overline{\mathrm{KE}}(20^{\circ}\mathrm{C})} = \frac{\frac{3}{2}\mathrm{k_b}\ (313)}{\frac{3}{2}\mathrm{k_b}\ (293)} = \frac{313}{293}$

(Here, $\mathrm{k_b}$ =Boltzmann's constant=$\frac{\mathrm{R}}{\mathrm{N_A}}$)

Q52

The number of Faraday's passed $= \frac{2\times10^{-3}\times16\times60}{96500}$

$= 1.99 \times 10^{-5}$

$\Rightarrow$ number of gram equivalent of Cu^{2+} deposited as $\mathrm{Cu(s)}$

$= 1.99 \times 10^{-5}$

$\Rightarrow$ number of moles of Cu^{2+} deposited

$= \frac{1.99}{2} \times 10^{-5} \approx 10^{-5}$

Absorbance is directly proportional to $\left[\mathrm{Cu}^{2+}\right]$.

No. of moles of Cu^{2+} initially present = 2 x 10^{-5} moles

$\left[\mathrm{Cu}^{2+}\right]_{\text{Initial}} = 2 \times 10^{-5} \times \frac{1000}{250}$

= 8 x10^{-5}M

Q53

Since the partial pressure reduces to half after

6.93 minutes, the half-life,$t_{1/2} = 6.93\text{min}$

$\mathrm{k} = \frac{0.693}{\mathrm{t}_{1/2}} = \frac{0.693}{6.93} = 0.10\,\mathrm{min}^{-1}$

For a gas phase first order reaction,

$\mathrm{k} = \frac{2.303}{\mathrm{t}}\log_{10}\left(\frac{\mathrm{p_i}}{\mathrm{p_A}}\right)$

$\mathrm{p_i} = 6.0\ \text{atm}\ ;\ \mathrm{p_A} = 6.0 - 4.0 = 2.0\ \text{atm}$

$\therefore\ \ 0.10 = \frac{2.303}{\mathrm{t}}\log_{10}\left(\frac{6.0}{2.0}\right)$

$\therefore \quad t = 23.03 \ \log \ (3.0) = 11.05 \ \min$

Q54

Ziegler-Natta catalyst is $TiCl_4 + Al(C_2H_5)_3$

Q55

Tertiary alcohols like 2-methylpropan-2-ol, 2, 4-dimethylpentan-2-ol react immediately with Lucas reagent $(conc. HCl \ + \ anhydrous \ ZnCl_2 \ )$ at room temperature.

Q56

Total count of geometrical isomers of square planar complex $[Mabcd]$ is 3 and $[Ma_4 \, b_2]$ is 2.

$\text{Total} = 5$

Q57

Mixtures (i), (ii) and (vi) behave as ideal solutions.

Q58

Chemical reaction is :

$5 \, Fe^{2+} + MnO_4 + 8H^+$

$\rightarrow Mn^{2+} + 4H_2O + 5 \, Fe^{3+}$

$\therefore \quad 5 \, mol Fe^{2x} \equiv 1 \, mol MnO_4$Now, 1000 mL of

100 mL of $0.1 \, M KM_n \, O_4 = 0.01 \, mol MnO_4$

$0.01 \, mol MnO_4 = 0.05 \, mol Fe^{2+}$

$= \ 0.05 \ mol \ FeSO_4$

$= 0.05 \times 152$

$= 7.6\ \text{g}\ FeSO_4$

Q59

	$3A \rightleftharpoons$	C	+	2D
Initially:	C			
at eq:	C−x	$\frac{x}{3}$		$\frac{2x}{3}$

Now, $\frac{2x}{3} = 2 \times (C - x)$

$x = 3C - 3x$

$C = \frac{4}{3}x$

$K_c = \frac{[C][D]^2}{[A]^3}$

$K_c = \frac{\frac{x}{3} \times \left(\frac{2x}{3}\right)^2}{\left(\frac{4}{3}x - x\right)^3} = \frac{\left(\frac{4}{27}\right)}{\left(\frac{1}{27}\right)} = 4$

Q60

$q = q_{AB} + q_{BC} + q_{CD} + q_{DA}$

$= nRT_B \cdot \ln \frac{2V_0}{V_0} + n \cdot C_{v,m} \cdot (T_C - T_B) + nRT_c \ln \frac{V_0}{2V_0} + n \cdot C_{v,m}(T_A - T_D)$

$= nR\ln 2(T_B - T_C) = 1 \times 2 \times 0.7 \times 200 = 280\,\text{cal/mol}$

Q61

$$\text{Given integral } = \int_0^1 0dx + \int_1^2 1\{x\}dx + \int_2^3 2^3\{x\}dx + \ldots\ldots + \int_9^{10} 9^3\{x\}dx$$

$$= \left(1^3 + 2^3 + \ldots\ldots 9^3\right) \int_0^1 \{x\}dx = \left(\frac{9 \times 10}{2}\right)^2 \times \int_0^1 xdx = \frac{2025}{2}$$

Q62

Any tangent to $y = x^2$ can be written as $y = mx - \frac{1}{4}m^2$ if it touches 2nd, then

$$mx - \frac{m^2}{4} = x^2 - 2x + 2$$

$$\Rightarrow \quad x^2 - (2+m)x + \left(2 + \frac{m^2}{4}\right) = 0$$

$$D = 0 \Rightarrow (m+2)^2 - 4\left(2 + \frac{m^2}{4}\right)$$

$$4m - 4 = 0 \Rightarrow m = 1 \Rightarrow 4x - 4y - 1 = 0$$

Q63

Given, $f(f(1)) = 0 \Rightarrow f(1 + \alpha + \beta) = 0$

$f(f(2)) = 0 \Rightarrow f(4 + 2\alpha + \beta) = 0$

Hence, roots of $f(x)$ are $\alpha + \beta + 1$ and $2\alpha + \beta + 4$

Now, sum of roots $= 3\alpha + 2\beta + 5 = -\alpha$

$\Rightarrow \quad 4\alpha + 2\beta = -5$

Now, product of roots $= (\alpha + \beta + 1)(2\alpha + \beta + 4) = \beta$

$(\alpha + \beta + 1)\frac{3}{2} = \beta$

From Equs. (1) and (2), $b = f(0) = \frac{-3}{2}$

Hence, $2|f(0)| = 3$

Q64

$$\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$$

$\Rightarrow \mathrm{a(b-1)(c-1)} - (1-a)(b-1) - (c-1)(1-a) = 0$

Dividing by $(1-a)(1-b)(1-c)$

$$\Rightarrow \frac{a}{1-a} + \frac{1}{1-c} + \frac{1}{1-b} = 0$$

$$\therefore \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$$

Q65

Given,

$$\text{Slope of normal } = \frac{-dx}{dy} = \frac{x^2}{xy - x^2y^2 - 1}$$

$x^2y^2dx + dx - xydx = x^2dy$

$x^2y^2dx + dx = x^2dy + xydx$

$x^2y^2dx + dx = x(xdy + ydx)$

$x^2y^2dx + dx = xd(xy)$

$$\frac{dx}{x} = \frac{dxy}{1 + x^2y^2}$$

$\ln kx = \tan^{-1}(xy)$ ... (i)

Curve passes though $(1, 1)$

So, In k $= \frac{\pi}{4} \Rightarrow$ k $= e^{\frac{\pi}{4}}$

Now from equation (i)

We get, $\frac{\pi}{4} + \ln x = \tan^{-1}(xy)$

$xy = \tan\left(\frac{\pi}{4} + \ell nx\right)$

$xy = \left(\frac{1-\tan(\ln x)}{1-\tan(\ln x)}\right)$...(ii)

Put $x = e$ in (ii)

We get, e. $y(e) = \frac{1+\tan 1}{1-\tan 1}$

Q66

$$(125)\ A + B = 3I \Rightarrow AA^T + BA^T = 3A^T \Rightarrow AA^T = 3A^T - BA^T \Rightarrow 4I = 3A^T - BA^T$$
$$\Rightarrow 12A^{-1} - BA^T + I = 5I$$
$$\therefore \det\left(12A^{-1} - BA^T + I\right) = 125$$

Q67

$S_n = {}^nC_0\,{}^nC_1 + {}^nC_1\,{}^nC_2 + \ldots + {}^nC_{n-1} \cdot {}^nC_n = {}^{2n}C_{n-1}$

$S_{n+1} = {}^{2n+2}C_n$

$$\frac{S_{n+1}}{S_n} = \frac{{}^{2n+2}C_n}{{}^{2n}C_{n-1}} = \frac{15}{4}$$

$$\Rightarrow \quad \frac{(2n+2)(2n+1)}{n(n+2)} = \frac{15}{4}$$

$$\Rightarrow \quad n^2 - 6n + 8 = 0$$

sum of roots $= 6$

Q68

$$\frac{1}{d}\left(\frac{a_2-a_1}{a_2a_1}+\frac{a_3-a_2}{a_3a_2}+\ldots.+\frac{a_{4001}-a_{4000}}{a_{4001}a_{4000}}\right)=10$$

$$=\frac{1}{d}\left(\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_2}-\frac{1}{a_3}+\ldots\ldots+\frac{1}{a_{4000}}-\frac{1}{a_{4001}}\right)=10$$

$$=\frac{1}{d}\left(\frac{1}{a_1}-\frac{1}{a_{4001}}\right)=10$$

$$=\frac{1}{d}\left(\frac{a_{4001}-a_1}{a_1\cdot a_{4001}}\right)=10$$

$$=\frac{4000}{a_1\cdot a_{4001}}=10 \quad (\text{ as } a_{4001}=a_1+4000d)$$

$$\Rightarrow \quad a_1a_{4001}=400; a_2+a_{4000}=50\Rightarrow (a_1+d)+(a_1+3999d)=50$$

$$(a_1-a_{4001})^2=(a_1+a_{4001})^2-4a_1a_{4001}$$

$$|a_1-a_{4001}|=30$$

Q69

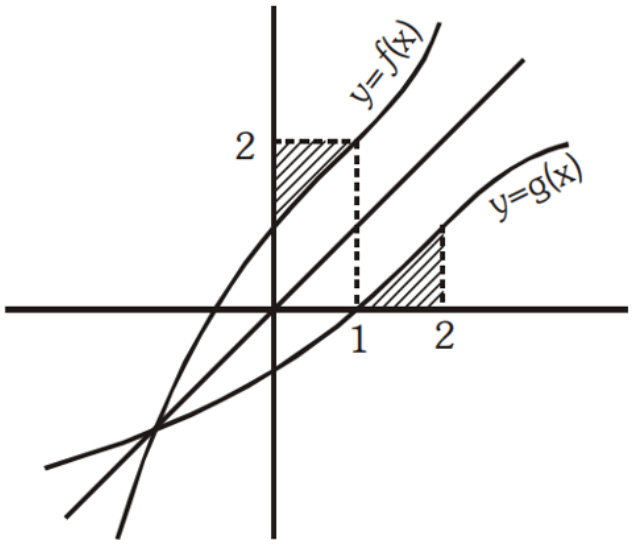
$f(x)=x^3-3x^2+3x+1$

$f'(x)=3x^2-6x+3=3\left(x^2-2x+1\right)$

$f'(x)=0 \quad \Rightarrow \quad x=1$

sign of $f'(x)$ does not change about $x=1$, then $x=1$ is point of inflection.

Area bounded by $y=g(x)$ and x-axis from $x=1$ to x $=2$ is same as area bounded by y $=f(\mathrm{x})$ and y-axis from y $=1$ to y $=2$.



Area of shaded region $=$ area of rectangle with 1 and 2 as sides - area bound by $f(x)$ from $x=0$ to $1=1/4$

Q70

$p \Rightarrow (p \vee \sim q) \equiv \sim p \vee (p \vee \sim q) \equiv (\sim p \vee p) \vee \sim q$

We know, $\sim p \vee p$ is always true

i.e. the given statement is always true, hence, a tautology

Q71

$$\underbrace{\int \left(\frac{1}{(\cos x)^{2019}} \right)}_{\text{I}} \cdot \underbrace{\operatorname{cosec}^2 x}_{\text{II}} dx - 2019 \int \frac{dx}{(\cos x)^{2019}}$$

$$\frac{1}{(\cos x)^{2019}} \cdot (-\cot x) + 2019 \int \frac{\sin x}{(\cos x)^{2020}} \cdot \cot x dx$$

$$\therefore \quad f(x) = \cot x; \quad g(x) = \cos x$$

$$\Rightarrow \quad \left| f\left(\frac{\pi}{4}\right) + g(0) \right| = 2$$

Q72

$$\begin{aligned}
&h(x) = g(x) + x \\
\Rightarrow &h'(x) = g'(x) + 1 \\
\Rightarrow &g'(x) = h'(x) - 1 \\
\Rightarrow &g''(x) = h''(x) \\
\Rightarrow &h''(x) - 3\left(h'(x) - 1\right) > 3 \\
\Rightarrow &h''(x) - 3h'(x) > 0 \\
\Rightarrow &\frac{d}{dx}\left(e^{-3x} h'(x)\right) > 0
\end{aligned}$$

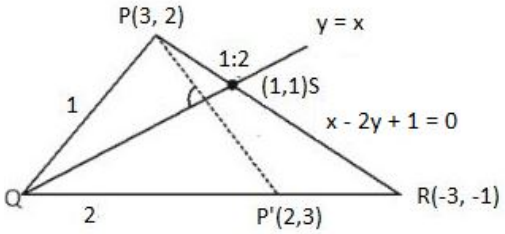
Let $P(x) = e^{-3x} h'(x)$

$$\begin{aligned}
\Rightarrow &P'(x) > 0 \\
\Rightarrow &P(x) \text{ is an increasing function.} \\
&P(0) = h'(0) = 0 \\
\Rightarrow &P(x) > 0 \forall x > 0 \\
\Rightarrow &h'(x) > 0 \forall x > 0 \\
\Rightarrow &h(x) \text{ is an increasing function } \forall x > 0
\end{aligned}$$

Q73

$|A| = 4, |B| = 6$, the number of one-one functions is ${}^6P_4 = 360$. The number of increasing functions is $\binom{6}{4} = 15$. The desired number is $360 - 15 = 345$

Q74



Point of intersection of $y = x$ & $x - 2y + 1 = 0$ is $S(1,1)$

Now, $PS : SR = PR : RQ = 1 : 2$

Hence, using section formula $R = (-3, -1)$

Let, reflection of P about the line $y = x$ is $P^{'}$

$\Rightarrow P^{'} = (2,3)$ which lies on the side QR

$\Rightarrow$ equation of QR is $4x - 5y + 7 = 0$

Point of intersection of QR & $y = x$ is

$(7,7) = Q$

Q75

Given,

$$\begin{aligned} 2\mathrm{x} - \mathrm{y} - \mathrm{z} &= -\mathrm{a} \\ \mathrm{x} - 2\mathrm{y} + \mathrm{z} &= \mathrm{b} \\ \mathrm{x} + \mathrm{y} - 2\mathrm{z} &= \mathrm{c} \end{aligned}$$

For consistent system of equations $\Delta \neq 0$ or $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$

$$\Delta = \begin{vmatrix} 2 & -1 & -1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{vmatrix} = 0$$

$$\Delta_1 = \begin{vmatrix} -\mathrm{a} & -1 & -1 \\ \mathrm{b} & -2 & 1 \\ \mathrm{c} & 1 & -2 \end{vmatrix} = 0$$

$\Rightarrow a+b+c=0$

$$\Delta_2=\begin{vmatrix}2 & -a & -1\\ 1 & b & 1\\ 1 & c & -2\end{vmatrix}=0$$

$\Rightarrow a+b+c=0$

$$\Delta_3=\begin{vmatrix}2 & -1 & -a\\ 1 & -2 & b\\ 1 & 1 & c\end{vmatrix}=0$$

$\Rightarrow a+b+c=0$

$$\left.\begin{matrix}a+b+c=0\\ 16a-4b+c=0\end{matrix}\right\}\Rightarrow \frac{a}{5}=\frac{b}{15}=\frac{c}{-20}$$

$\Rightarrow a=\lambda$

$b=3\lambda$

$c=-4\lambda$

$\Rightarrow$ Sum of roots $=\frac{-b}{a}=-3$

Q76

Given, $f(x+y)=f(x)+f(y)-xy-1, \forall x, y \in R$

$\therefore f(x+1)=f(x)+f(1)-x-1[\text{ putting } y=1]$

$\Rightarrow f(x+1)=f(x)-x[\because f(1)=1]$ AARKS

$\therefore f(n+1)=f(n)-n<f(n)$

$\Rightarrow f(n+1)<f(n)$

So, $f(n)<f(n-1)<f(n-2)<\ldots<f(3)<f(2)<f(1)=1$

$\therefore f(n)=n$ holds only for $n=1$

Q77

$$T_n=\tan^{-1}\left(\frac{4}{4n^2+3}\right)=\tan^{-1}\left(\frac{1}{n^2+(3/4)}\right)=\tan^{-1}\left(\frac{\left(n+\frac{1}{2}\right)-\left(n-\frac{1}{2}\right)}{1+\left(n+\frac{1}{2}\right)\left(n-\frac{1}{2}\right)}\right)$$

$$T_n=\tan^{-1}\left(n+\frac{1}{2}\right)-\tan^{-1}\left(n-\frac{1}{2}\right)$$

$$S_n=\tan^{-1}\left(n+\frac{1}{2}\right)-\tan^{-1}\left(\frac{1}{2}\right)\Rightarrow S_\infty=\frac{\pi}{2}-\tan^{-1}\left(\frac{1}{2}\right)$$

Q78

Given, $R = \{(x, y) \in W \times W$: the word x and y have atleast one letter in common$\}$

Let, $W = \{\text{cat},\ \text{toy},\ \text{you}, \ldots\}$

Clearly, R is reflexive and symmetric but not transitive.

[Since, cat R toy, toy Ryou $\nRightarrow$ cat R you]

Q79

$$L = \lim_{x \to \infty} \sqrt{n} \int_0^1 \frac{dx}{(1+x^2)^n}$$

$$(1+x^2)^n = 1 + nx^2 + \ldots..$$

$$\therefore \quad (1+x^2)^n > 1 + nx^2$$

$$\frac{1}{(1+x^2)^n} < \frac{1}{1+nx^2}$$

$$\therefore \quad \int_0^1 \frac{dx}{(1+x^2)^n} < \int_0^1 \frac{dx}{(1+x^2)}$$

$$= \frac{1}{n} \int_0^1 \frac{dx}{\frac{1}{n} + x^2} = \frac{1}{n} \sqrt{n} \left(\tan^{-1} x\sqrt{n}\right)_0^1$$

$$= \frac{1}{\sqrt{n}} \tan^{-1} \sqrt{n}$$

$$\therefore \quad \lim_{x \to \infty} \sqrt{n} \int_0^1 \frac{dx}{(1+x^2)^n} < \lim_{x \to \infty} \tan^{-1} \sqrt{n}$$

$$\therefore \quad L < \frac{\pi}{2}$$

$$\therefore \quad L < 2$$

Q80

Here, $\lim_{x \to 0} \frac{x \sin(\sin x) - \sin^2 x}{x^6}$

$$= \lim_{t \to 0} \frac{\sin^{-1}(t) \cdot \sin t - t^2}{\left(\sin^{-1} t\right)^6}$$

$$= \lim_{t \to 0} \frac{\sin^{-1}(t) \cdot \sin t - t^2}{t^6} \cdot \frac{t^6}{\left(\sin^{-1} t\right)^6}$$

[put $\sin x = t$]

$$= \lim_{t \to 0} \frac{\left\{t + \frac{t^3}{6} + \frac{9t^5}{120} + \frac{5t^7}{112} + \ldots\right\} \cdot \left\{t - \frac{t^3}{3!} + \frac{t^5}{5!} - \ldots\right\} - t^2}{t^6} \times 1$$

$$= \lim_{t\to 0} \frac{t^6\left(\frac{1}{5!}-\frac{1}{6\cdot(3!)}+\frac{9}{120}\right)+\text{higher powers to } t}{t^6}$$

$$= \frac{1}{120}+\frac{9}{120}-\frac{1}{36}=\frac{1}{18}$$

Q81

Point of intersection of the lines is $(0,1,1)$ which lies on the plane P.

D.r. of L_1 is $(2,2,1)$& D.r. of L_2 is $(1,1,-1) \Rightarrow$ a vector parallel to P is

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ 1 & 1 & -1 \end{vmatrix} = -3\hat{i}+3\hat{j}$$

$$\therefore \quad \text{Normal to the plane } = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 3 & 0 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= 3\hat{i}-3\hat{j}-6\hat{k}$$

$\Rightarrow$ Equation of plane is

$$1.\ (x-0)+(y-1)+2(z-1)=0$$

$$\Rightarrow \quad x+y+2z=3$$

$$\because \quad (a,-2,0) \text{ lies on } P \Rightarrow \quad a=5$$

Q82

Let $z=x+iy$

S_1 denotes the interior of circle of radius 4 units

S_3 denotes $x>0$

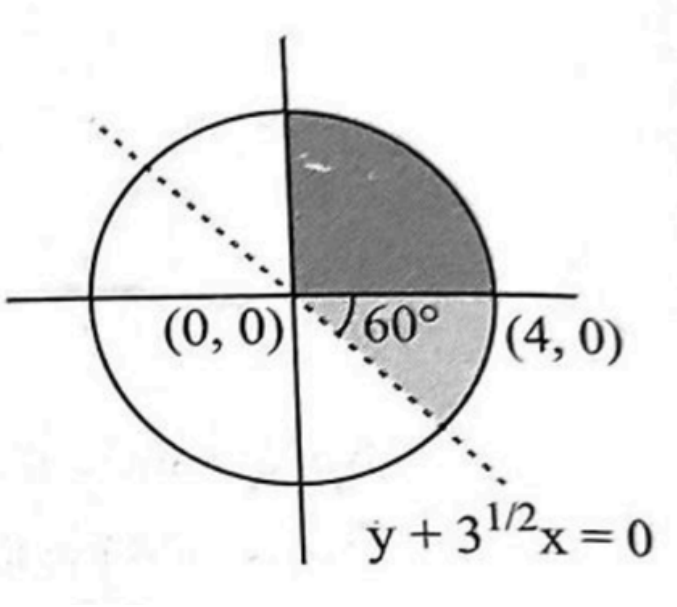
$$S_2 = \text{Im}\left(\frac{(x-1+i(y+\sqrt{3}))(1+i\sqrt{3})}{4}\right) > 0$$

$$S_2 = i(y+\sqrt{3}x) > 0$$

Now the shaded region represents the required area

Required area = Area of quarter of circle + Area of sector

$= \frac{\pi r^2}{4} + \frac{\pi r^2 \theta}{360^\circ} = 4\pi + \frac{8}{3}\pi \Rightarrow \frac{20}{3}\pi$



Q83

$$V_1 = (\vec{a} + \vec{b}) \cdot ((\vec{b} + \vec{c}) \times (\vec{c} + \vec{a})) = 2[\vec{a}\,\vec{b}\,\vec{c}]$$

$$V_2 = \frac{1}{6}(\vec{a} \times \vec{b}) \cdot [(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})]$$

$$= \frac{1}{6}(\vec{a} \times \vec{b}) \cdot [((\vec{b} \times \vec{c}) \cdot \vec{a})\vec{c} - ((\vec{b} \times \vec{c}) \cdot \vec{c})\vec{a}]$$

$$= \frac{1}{6}[\vec{a}\vec{b}\vec{c}]^2$$

$$[\vec{a}\vec{b}\vec{c}] = \begin{vmatrix} -3 & 1 & 1 \\ 4 & 2 & 4 \\ 2 & 2 & 0 \end{vmatrix} = 36$$

$\therefore V_1 = 72$ and $V_2 = 216$

Q84

$\bar{x} = \frac{\sum x_1}{50} = 50$

$\Rightarrow \sum x_i = 2500$

Correct $\sum x_i = 2500 - 99 = 2401$

Correct $\bar{x} = \frac{2401}{49} = 49$

$\sigma^2 = \frac{\sum x_i^2}{n} - \bar{x}^2$

$\Rightarrow 100 = \frac{\sum x_i^2}{50} - 2500$

$\Rightarrow \sum x_i^2 = 130000$

Correct $\sum x_i^2 = 130000 - 99^2 = 120199$

Correct $\sigma^2 = \frac{\sum x_i^2}{n} - \overline{x}^2$

$= \frac{120199}{49} - (49)^2$

$= \frac{120199 - 117649}{49} = \frac{2550}{49} = \frac{\lambda}{49}$

Hence, $\lambda = 2550$

Q85

$h(x) = x^2$ for $x \geq \frac{1+\sqrt{5}}{2}$

$h(x) = 1 + x$ for $0 \leq x \leq \frac{1+\sqrt{5}}{2}$

$h(x) = 1 - x$ for $\frac{-81}{100} \leq x < 0$

$h(x) = 1 - x^2$ for $x < \frac{-81}{100}$

h is non-continuous at $\frac{-81}{100}$

h is non-differentiable at $\frac{1+\sqrt{5}}{2}$, 0 and $\frac{-81}{100}$ $m = 3$ and $n = 1$

Hence, $m + n = 4$

Q86

Let A be the event that the letter is from TATANAGAR and B be the event that letter is from CALCUTTA.

Also, let E be the event that on the letter, two consecutive letters TA are visible.

$P(A) = \frac{1}{2}; \quad P(B) = \frac{1}{2}$

And $P(E/A) = \frac{2}{8}$ and $P(E/B) = \frac{1}{7}$

[If the letter is TATANAGAR, we see that the events of two consecutive letters visible are TA, AT, TA, AN, NA, AG, GA, AR]

So, $\quad P(E/A) = \frac{2}{8}$ and same in case of CALCUTTA, so $P(E/B) = \frac{1}{7}$

Therefore, $P(A/E) = \frac{P(A)P(E/A)}{P(A)P(E/A)+P(B)P(E/B)}$

$= \frac{(1/2)(2/8)}{[(1/2)(2/8)]+[(1/2)(1/7)]} = \frac{(1/8)}{(1/8)+(1/14)} = \frac{(1/8)}{(11/56)} = \frac{7}{11}$

$p = 7, q = 11, p + q = 11 + 7 = 18$

Q87

$x = -1$ is the root of the quadratic equation $ax^2 + 2bx + c = 0$

$\Rightarrow a - 2b + c = 0 \Rightarrow 2b = a + c$

$\Rightarrow a$ and c both must be even or odd.

So, the total number of quadratic polynomials, $k = {}^{1001}\mathrm{C}_2 \times 2 + {}^{1001}\mathrm{C}_2 \times 2$

$= {}^{1001}\mathrm{C}_2 \times 2 \times 2 = \frac{1001\times1000}{2} \times 2 \times 2$

$\Rightarrow \frac{k}{10^3} = 2002$

Q88

$B^2 = I$

$$AB = \begin{bmatrix} a & x & p \\ y & q & b \\ r & c & z \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} p & x & a \\ b & q & y \\ z & c & r \end{bmatrix}$$

$$AB = AB^3 = \ldots\ldots = AB^{19} = \begin{bmatrix} p & x & a \\ b & q & y \\ z & c & r \end{bmatrix}$$

$\text{tr.}\left(AB + AB^3 + \ldots\ldots + AB^{19}\right) = 210$

$\Rightarrow 10(p + q + r) = 210 \Rightarrow p + q + r = 21, p, q, r \in N$

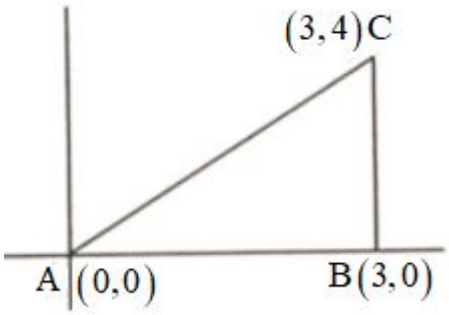
$p' + q' + r' = 18, p', q', r' \in W$

$\therefore$ Number of ordered triplets $(p, q, r) = {}^{20}C_2 = \frac{20\times19}{2} = 190$

Q89

$$f(x) = x^4 - 4x^3 - 8x^2 + a$$
$$f'(x) = 4\left(x^3 - 3x^2 - 4x\right)$$
$$= 4x\left(x^2 - 3x - 4\right)$$
$$= 4x(x-4)(x+1) = 0 \text{ at } x = -1, 0, 4$$
$$f(-1) = a - 3 \leq 0, a \leq 3$$
$$f(0) \geq 0 \Rightarrow a \geq 0$$
$$a \in [0, 3]$$
$$\text{Sum } = 0 + 1 + 2 + 3 = 6$$

Q90



$AC = 5 = 2ae$

$\Rightarrow 2ae = 5 \ldots (\text{i})$

Also $AB + BC = 2a$

$\Rightarrow 2a = 7 \Rightarrow a = \frac{7}{2}$

$\therefore e = \frac{5}{7}$

$\Rightarrow 1 - \frac{b^2}{a^2} = \frac{25}{49}$

$\Rightarrow b = \sqrt{6}$

Length of the latus rectum, $\frac{2b^2}{a} = \frac{2\times 6}{\frac{7}{2}}$ units

$\Rightarrow p = \frac{7}{2} = 3.5$